Trading Scope for Credibility in Difference-in-Differences

Honest Inference After Selecting Cohorts on Pre-Trends

Parush Arora* Abhishek Chand†

Abstract

In staggered difference-in-differences, parallel trends may hold for some treated cohorts and fail for others, so the average treatment effect on the treated (ATT), an average over all cohorts including the offenders, is biased and its honest confidence set is wide. Dropping the offending cohorts and reporting the effect for the credible ones—the difference-in-differences analogue of the local average treatment effect and of overlap weighting—point-identifies the sub-aggregate under a weaker parallel-trends requirement that can hold when the ATT's fails (Proposition 1). But when the credible cohorts are read off the observed pre-trends, selection reintroduces the pre-test problem the honest-DiD literature was built to avoid, and existing honest methods do not cover it, because they never select. Our contribution is inference for this selected target: we compose the post-selection carving of Lee et al. (2016) with the sensitivity bounds of Rambachan and Roth (2023) into a single interval, uniformly valid against *both* the data-driven selection and the residual violation the screen cannot rule out, with no separation or consistent-selection condition. We characterize in closed form when narrowing the target lowers risk—a feasible rule, reported alongside both honest intervals, that can cost efficiency but never coverage. In an application, a significantly positive pooled estimate of the shale boom's effect on local house prices proves to rest on cohorts already trending before onset; the credible subpopulation—the counties whose oil-and-gas onset did not coincide with the mid-2000s housing run-up—reveals no effect.

Keywords: Difference-in-differences, Staggered adoption, Parallel trends, Robust inference, Sensitivity analysis, Partial identification.

JEL classification: C1, C23, C51.

*Department of Economics, Ashoka University.
†Independent researcher.

1 Introduction

Difference-in-differences is among the most widely used research designs in empirical economics, and its credibility rests on the parallel trends assumption, that absent treatment treated and comparison units would have evolved in parallel. Researchers assess it by testing for pre-treatment differences in trends, but that practice is fragile. Pre-trends tests are often underpowered against economically meaningful violations (Bilinski and Hatfield, 2020; Freyaldenhoven et al., 2019; Kahn-Lang and Lang, 2020; Roth, 2022), conditioning on passing them induces a pre-test selection bias (Roth, 2022), and even a clean pre-period does not guarantee a clean post-period, since nothing prevents a confound from arriving at the moment of treatment (Kahn-Lang and Lang, 2020).

A growing literature, surveyed in Roth et al. (2023), responds by relaxing parallel trends while holding the ATT fixed as the target. Building on Manski and Pepper (2018), Rambachan and Roth (2023) bound how far post-treatment violations can depart from the observed pre-trends and report a uniformly valid confidence set for the ATT; Kwon and Roth (2024) sharpen the set with an empirical-Bayes prior on the violation; and de Chaisemartin (2026) rank the post-treatment differences against the pre-treatment ones. All keep the ATT and use the pre-trends to extrapolate the violation across periods. But the ATT is where a structural difficulty lies. When parallel trends fails for some cohorts and holds for others, the ATT is an average over all treated cohorts, including the offenders: its point estimate is biased, and its honest confidence set, though valid, is wide, because it must accommodate the worst cohort in the average.

The target change is the easy half and an old idea; the work is that selecting the credible cohorts from the pre-trends revives the very pre-test bias honest-DiD exists to avoid, and no existing honest method addresses it, because none selects. This paper supplies that inference. Applied practice already retreats to the part of a design it trusts, but piecemeal and without inference to match: researchers exclude treated units whose inclusion would contaminate the comparison, select and weight comparison units for pre-treatment fit (Abadie et al., 2010), and, less formally, adjust samples and specifications until the pre-trends look flat—the specification search whose statistical costs Roth (2022) documents. Each is a retreat to a more credible subpopulation, made ad hoc or on design grounds, and none carries inference valid against the selection it performs. When the population ATT is not credibly identified, we make that retreat explicit and give it a target and an inference: we drop to the subpopulation of cohorts where identification is credible—the logic of the local average treatment effect (Imbens and Angrist, 1994), of the optimal-subpopulation average treatment effect under limited overlap (Crump et al., 2009), and of overlap weighting (Li et al., 2018), each of which trades the original estimand for a credibly identified effect on a selected subpopulation. The credible-subpopulation LATT is a reweighting of standard group-time effects (Callaway and Sant'Anna, 2021) over the cohorts whose parallel trends is credible.[1] Credibility can

[1]These estimators (Callaway and Sant'Anna, 2021; Sun and Abraham, 2021; Borusyak et al., 2024; de Chaisemartin and D'Haultfœuille, 2020) repair the aggregation failure of two-way fixed effects under staggered adoption (Goodman-Bacon, 2021), whose implicit weights can turn negative under treatment-effect heterogeneity. That repair concerns aggregation, not identification, since they maintain parallel trends and are biased when it fails, which is the failure we address.

be decided ex ante, from covariates or institutional knowledge, but in the leading case it is read from the pre-trends themselves—and selecting on the same pre-trends used for estimation is exactly the step the honest-DiD literature avoided, because it revives the pre-test bias that motivates that literature. Our central contribution is to make that selection honest: to give inference for the LATT that is uniformly valid against both the data-driven selection and the residual violation of parallel trends that a pre-trend screen cannot rule out.

That the LATT is point-identified under parallel trends on the selected cohorts alone—a weaker requirement that can hold when the ATT's fails (Proposition 1)—is immediate, and we treat it as a benchmark rather than a contribution: the screen selects on pre-trends, and a flat pre-trend does not certify the flat post-trend that identification requires, so the honest interval, not the point, is the object we carry. The paper's work is the inference that a data-driven selection demands, and it makes three contributions. *First, and centrally, honest post-selection inference.* We combine the post-selection "carving" of Lee et al. (2016) with the sensitivity bounds of Rambachan and Roth (2023) into a single interval for the LATT that is uniformly valid in the data-generating parameter, with no separation or consistent-selection condition (Proposition 6): the carved component absorbs the data-driven selection and the level bound absorbs the residual identification failure, and the two compose. Neither ingredient suffices alone. Post-selection carving (Lee et al., 2016) delivers valid inference when the target is point-identified conditional on the selected set; here it is not, because a flat pre-trend does not certify the flat post-trend that identification requires, so carving alone leaves a residual violation it cannot touch, and composing it with a sensitivity class is what restores honesty. This is the step the fixed-target sensitivity analysis never takes, because it never selects, and it carries selection and identification uncertainty on separate instruments. *Second, a feasible decision rule.* We characterize in closed form when the trade pays—when the point-identified LATT dominates the set-valued ATT. A width-dominance theorem gives the explicit threshold on the dropped cohorts' violation above which the credible-subpopulation honest interval is strictly shorter than the Rambachan and Roth (2023) interval for the ATT, a threshold estimable from the estimated covariance alone (Theorem 1); and we turn the characterization into an estimable rule reported alongside both honest intervals (Proposition 8). Because both intervals are honest, the rule can never cost coverage, only efficiency; under homogeneous effects it selects the lower-risk procedure with probability approaching one, and under heterogeneity it returns a breakdown value Γ^* in the idiom of a sensitivity analysis. *Third, evidence.* Simulations trace the method's advantage across the informativeness of pre-trends, and an application to the shale boom withdraws a spuriously positive pooled estimate of the effect on house prices, locating it in the cohorts already trending before onset. An appendix develops the optimal credible weighting that the same sensitivity class implies (Proposition 9)—a soft-thresholded weighting of cohorts by credibility net of noise, of which the flatness screen is a special case, completing the analogy to the variance-minimizing overlap weights of Crump et al. (2009).

Table 1 places the paper among existing robust-DiD methods. Where they hold the ATT fixed and use pre-trends to bound or extrapolate the post-treatment violation, we change the target

Table 1: Where this paper sits. Existing robust-DiD methods hold the ATT fixed and use pre-trends to bound or extrapolate the post-treatment violation; none selects, so none confronts selection in its inference. Our contribution is the last column: an interval uniformly valid against the data-driven selection *and* the residual violation, a guarantee the fixed-target methods have no need for.

Approach	Target	Use of pre-trends	Inference guarantee
Pre-trends test, then point DiD	ATT	Screen; keep design if the test passes	None post-test (pre-test bias)
Rambachan and Roth (2023)	ATT	Bound post- by pre-treatment violation (Δ)	Uniform over $\delta \in \Delta$
Kwon and Roth (2024)	ATT	Calibrate an (empirical-)Bayes prior on δ	Bayesian credible set
de Chaisemartin (2026)	ATT	Rank post- against pre-treatment DiDs	Valid under the ranking
This paper	**LATT** (credible subpop.)	Select credible cohorts; bound residual violation	**Uniform vs. selection *and* residual violation**

and treat the resulting selection as the object requiring inference; we nest the sensitivity analysis of Rambachan and Roth (2023) by applying it to the retained sub-aggregate, and we connect to the post-selection inference literature (Lee et al., 2016; Leeb and Pötscher, 2005, 2008) that the honest-DiD papers, by never selecting, did not need.

The remainder of the paper is organized as follows. Section 2 fixes the staggered-DiD setup, defines the credible-subpopulation LATT, and develops its identification, estimation, honest inference, and the decision rule. Section 3 reports the simulation study, Section 4 applies the method to the local house-price effect of the shale boom, and Section 5 concludes. The procedure, assembled in one place, is the following.

The credible-subpopulation procedure.

1. *Estimate* cohort-level (group-time) event studies with any heterogeneity-robust estimator (Callaway and Sant'Anna, 2021; Sun and Abraham, 2021; Borusyak et al., 2024; de Chaisemartin and D'Haultfœuille, 2020).

2. *Select* the credible cohorts—from institutional knowledge or a covariate split (ex ante), or by screening the pre-trends for flatness (data-driven).

3. *Estimate* the LATT by reweighting the retained cohorts' post-treatment effects.

4. *Report honest inference*: the level-bound sensitivity interval on the retained set, carved for the selection when it is data-driven (Proposition 6), and the breakdown violation M^*.

5. *Report the trade*: both honest intervals (LATT and ATT) with the switching diagnostic ($\hat{D}$, $\widehat{\Delta\mathrm{Var}}$, Γ^*) that signals which is more informative (Proposition 8).

2 The credible-subpopulation LATT

2.1 Setup

We adopt the staggered adoption framework of Callaway and Sant'Anna (2021). There are periods $t = 1, \ldots, T$ and units i indexed by their first treatment date $G_i \in \{2, \ldots, T\} \cup \{\infty\}$, where $G_i = \infty$ denotes never-treated. Let $Y_{it}(g)$ denote the potential outcome in period t if unit i is first treated in period g, and $Y_{it}(\infty)$ the never-treated potential outcome. The building-block causal parameter is the group-time average treatment effect on the treated,

$$\mathrm{ATT}(g, t) = \mathbb{E}[\, Y_{it}(g) - Y_{it}(\infty) \mid G_i = g\,]\,. \tag{1}$$

Summary parameters are weighted aggregations $\theta = \sum_g \sum_t w(g,t)\, \mathrm{ATT}(g,t)$ with researcher-chosen weights (Callaway and Sant'Anna, 2021), of which the overall ATT and the event-study path are special cases.

It is convenient to work with the event-study representation and the decomposition of Rambachan and Roth (2023). Let $\hat\beta = (\hat\beta'_{\mathrm{pre}}, \hat\beta'_{\mathrm{post}})'$ be an asymptotically normal vector of event-study coefficients estimated by any of the heterogeneity-robust procedures above, with $\sqrt{n}(\hat\beta - \beta) \to \mathcal{N}(0, \Sigma)$. The estimand decomposes as

$$\beta = \tau + \delta, \qquad \tau_{\mathrm{pre}} = 0, \tag{2}$$

where τ collects the causal effects (zero before treatment, by no anticipation) and δ is the differential trend between treated and comparison groups that would have occurred absent treatment. Parallel trends is the restriction $\delta_{\mathrm{post}} = 0$, under which $\beta_{\mathrm{post}} = \tau_{\mathrm{post}}$, and a pre-trends test is a test of $\delta_{\mathrm{pre}} = 0$.

We will require the analogous objects at the cohort level. Write δ_g for the differential trend of cohort g and say that parallel trends holds for g if $\delta_{g,\mathrm{post}} = 0$. A key feature of applications is that this may hold for some cohorts and fail for others. Defending the timing of treatment as unrelated to the outcome path is already delicate in a two-group design, and staggered adoption raises the burden rather than lowering it, since each cohort's timing must be defended on its own, one cohort's adoption perhaps as good as randomly timed while another's coincides with a confounding shock.

Finally, we distinguish two regimes for how cohorts are judged credible, because they carry different inferential guarantees. Under ex-ante (or independent) selection, credibility is decided from pre-determined covariates, institutional knowledge, or an independent data split, so that the selected set is a fixed S statistically independent of the estimation errors in $\hat\beta$. Under data-driven selection, credibility is assessed from the realized pre-trends $\hat\beta_{\mathrm{pre}}$, so that the selected set $\hat S$ is random and depends on the estimation sample, with a deterministic population counterpart S^*, the set a population pre-trend statistic would select. The identification argument below holds for any set held fixed, and so identifies the random target $\theta_{\hat S}$ conditional on the realized $\hat S$. Identifying the fixed population parameter θ_{S^*} instead requires $\hat S$ to recover S^*, a selection-consistency condition we return to under inference (Proposition 5). The distinction is thus minor for identification but central for inference, and we treat it explicitly below.

2.2 Estimand and identification

Let S denote a set of cohorts judged to have credible parallel trends, produced by a selection rule R that we specify below. We define the target as the treatment effect for that subpopulation.

Definition 1 (Credible-subpopulation LATT)**.** *For cohort weights $w_g > 0$ (e.g. proportional to cohort size) and a within-cohort aggregation $a(\cdot)$, a linear functional of a cohort's post-treatment vector (a fixed event-time, selecting one coordinate, or an average over post-treatment periods), the credible-subpopulation local ATT, defined for any nonempty S, is*

$$\theta_S \;=\; \frac{\sum_{g\in S} w_g\,\mathrm{ATT}(g,\cdot)}{\sum_{g\in S} w_g}, \qquad \mathrm{ATT}(g,\cdot) \;=\; a(\tau_{g,\mathrm{post}})\,, \tag{3}$$

the aggregated causal effect of cohort g.

The target is identified under a weaker condition than the ATT—a benchmark we state before turning, in Section 2.4, to the honest interval that is the paper's actual object.

Proposition 1 (Identification benchmark)**.** *If parallel trends holds for every $g \in S$, i.e. $\delta_{g,\mathrm{post}} = 0$ for all $g \in S$, then θ_S is point-identified by the reweighted aggregated group-time effects, $\theta_S = \big(\sum_{g\in S} w_g\, a(\beta_{g,\mathrm{post}})\big)/\sum_{g\in S} w_g$, and this holds irrespective of whether $\delta_{g,\mathrm{post}} \neq 0$ for cohorts $g \notin S$.*

Proof. For $g \in S$, $\beta_{g,\mathrm{post}} = \tau_{g,\mathrm{post}} + \delta_{g,\mathrm{post}} = \tau_{g,\mathrm{post}}$ by the hypothesis and (2), so $a(\beta_{g,\mathrm{post}}) = a(\tau_{g,\mathrm{post}}) = \mathrm{ATT}(g,\cdot)$ by linearity of a. The reweighted sum over S therefore equals the weighted average of the aggregated causal effects, which is θ_S by definition. Cohorts outside S receive zero weight, so their differential trends do not enter. $\square$

Proposition 1 is elementary, and deliberately so. The paper's work is not in the identification, which merely restates that a weighted average over cohorts where parallel trends holds is identified, but in the honest inference for the data-driven selection this target invites, developed in Section 2.4. Changing the target to the credible subpopulation weakens the identifying requirement from parallel trends on all cohorts to parallel trends on S, which can hold when the former fails, at the cost of a narrower question whose subpopulation S must then be characterized. The weakening is genuine but conditional, and two things must be kept separate. The identifying content is the substantive restriction $\delta_{g,\mathrm{post}} = 0$ for $g \in S$, an assumption about post-treatment trends. The flatness screen of Section 2.3 is a statistical classification rule that supplies evidence for that restriction from the pre-period, not the restriction itself, since a cohort flat before treatment may still drift after it. Point identification of the causal θ_S thus holds only when the restriction does, and the screen cannot certify it. We therefore treat the point-identified θ_S as a benchmark and pair it with the honest sensitivity bounds developed below, which bound the residual post-treatment violation the screen leaves behind and, in doing so, return a set rather than a point when that violation is nonzero.

The narrower question carries an interpretability cost that we state plainly rather than leave for the reader to raise. The LATT is our target, and against it the estimator is honestly centered; the mean-squared-error comparison of Section 2.5, which charges the LATT a γ^2 term, treats the ATT as the estimand and is offered as a courtesy to a reader who insists on it, not as a concession that the LATT is a biased estimate of something else. What must be conceded is different: the credible subpopulation is delimited by a statistical property—flat, credible pre-trends—rather than by a substantive behavioral criterion, so it is less transparently interpretable than the compliers of the local average treatment effect (Imbens and Angrist, 1994), and when treatment effects are heterogeneous it answers a different and possibly less policy-relevant question than the ATT for the full treated population. That gap is exactly the composition gap Γ of Section 2.5, unidentified because it depends on the effects of the cohorts the screen discards. We do not argue it away: we report both honest intervals with the breakdown value Γ^*, so the reader sees the full-population target alongside the credible one and how far the discarded cohorts' effects must differ to overturn the reading. Where the retained set admits a substantive description—in the application of Section 4, the counties whose oil-and-gas onset did not coincide with the mid-2000s housing run-up—we recommend leading with it, since it converts a cohort selected for a convenient statistical property into a subpopulation a reader can name.

The construction uses staggered timing only to supply the groups. What it requires is a partition of the treated population into groups whose parallel trends can each be assessed. Adoption cohorts are the natural groups, and their credibility heterogeneity comes for free from the variation in when and why units adopt. In a non-staggered design the same construction applies with groups defined by covariates or geography, provided the groups differ in credibility and there are enough pre-periods to estimate each group's pre-trend rather than chase noise. The groups should accordingly be subpopulations large enough to assess, not individual units, whose pre-trends are too noisy to select on without reviving the pre-test bias in its most severe form.

2.3 Selection and estimation

The selection rule R maps the data to the set S, and its form fixes both which subpopulation θ_S describes and the inferential guarantees available. It takes two forms, matching the two regimes of Section 2.1. Under ex-ante selection, S is fixed before estimation from pre-determined covariates, institutional knowledge of which cohorts were plausibly confounded, or an independent data split. Under data-driven selection, credibility is read from the estimated pre-trends, and a cohort enters S when its estimated pre-trend is close to flat,

$$S \;=\; \big\{\, g : \max_{e<0} |\hat{\beta}_{g,\mathrm{pre}}(e)| \leq c \,\big\}, \tag{4}$$

for a threshold c.

This is the direct reading of parallel trends. The assumption is that the treated-comparison difference would have stayed constant absent treatment, so a cohort is credible when its pre-treatment difference is close to constant, and the screen keeps exactly those. It is posed in the same currency as the honest inference of the next subsection, which bounds how far the post-treatment difference could have wandered from flat, so selection and inference test one object, the level of the parallel-trends violation.[2]

The threshold trades scope against credibility, with a lax c retaining more cohorts but admitting larger residual violations and a strict c the reverse. Because rule (4) reads S off the noisy estimates $\hat{\beta}_{\text{pre}}$, the selected set is random and dependent on the estimation sample, which is the source of the pre-test bias below and of the inference subtleties that follow it.

We do not recommend committing to a single c. The sensitivity interval of Section 2 is applied to whichever set c selects, and its bound M must dominate that set's residual *post*-treatment violation, so the effect of c on the required M runs through how far screening on flat pre-trends also flattens post-trends. When pre-trends are informative about post-treatment violations, a stricter c yields a cleaner set that a smaller M suffices to cover, while a laxer c buys scope at the cost of a larger M, so the threshold trades scope for precision rather than credibility for nothing. When they are not, tightening c need not shrink the post-treatment violation and the required M can be nonmonotone in c. Either way the honest object to report is the estimand together with its interval traced as a function of c, the selection analogue of the breakdown curve we report for M. A reader then sees the scope-credibility-precision frontier directly and need not trust a single cutoff or a presumed monotone mapping. When a single value is nonetheless wanted, we recommend calibrating it to negligibility rather than to significance, setting c to the largest pre-trend that would bias θ_S by less than a stated tolerance, in the spirit of the non-inferiority tests of Bilinski and Hatfield (2020), rather than to a critical value of a pre-trends test, which would reinherit the low power that motivates the paper. The application of Section 4 reports such a path.

The estimator $\hat{\theta}_S$ simply reweights, over the selected cohorts, the sample group-time effects of whichever heterogeneity-robust estimator one uses, and nothing below depends on that choice.

Proposition 2 (Consistency and efficiency)**.** *Under ex-ante selection and the regularity conditions of the underlying group-time estimator, $\hat{\theta}_S$ is consistent for θ_S. If in addition parallel trends holds for all cohorts, $\hat{\theta}_S$ is consistent for the ATT when either treatment effects are homogeneous or S comprises all cohorts, and otherwise for the selected-cohort LATT, which differs from the ATT by the gap between the selected- and full-population weighted averages of the cohort effects. In either case there is an efficiency loss relative to estimators that use all cohorts.*

The consistency clause of Proposition 2 invites a sharper question: when does narrowing the target cost nothing, in that the LATT and the ATT are the same object? The answer separates equivalence of the two targets from convergence of the two estimators.

[2] A researcher willing to trust that a straight pre-trend would have continued linearly can relax flatness to approximate linearity, screening instead on the pre-trend's curvature (its second difference) and pairing it with the smoothness sensitivity class of Rambachan and Roth (2023) and a linear-extrapolation estimator (Armstrong and Kolesár, 2018). This buys the steep-but-straight cohorts that flatness discards, at the cost of the extrapolation assumption. The principle is the same in either currency. The screen should be posed in whichever functional the sensitivity class bounds, flatness with the level bound and curvature with the smoothness class, since level and curvature rank cohorts differently and select different sets.

Proposition 3 (Coincidence of the LATT and the ATT)**.** *Write $\theta_g := \mathrm{ATT}(g, \cdot)$ for the cohort-level causal effect, $\mathcal{G}$ for the full set of treated cohorts, and $\theta_{\mathcal{G}}$ for the ATT, the aggregate over all of $\mathcal{G}$. For a fixed selected set S the two targets coincide, $\theta_S = \theta_{\mathcal{G}}$, if and only if the retained cohorts' weighted-average effect equals the full population's,*

$$\frac{\sum_{g \in S} w_g \, \theta_g}{\sum_{g \in S} w_g} = \frac{\sum_{g \in \mathcal{G}} w_g \, \theta_g}{\sum_{g \in \mathcal{G}} w_g}.$$

This holds in particular when no cohort is dropped, $S = \mathcal{G}$, and when effects are homogeneous, $\theta_g \equiv \theta$, under which every subpopulation shares the common value; and more generally whenever selection is uninformative about effect size, in that the dropped cohorts $\mathcal{G} \setminus S$ carry the same weighted-average effect as the retained ones. Under data-driven selection the two estimators *converge, $\hat{\theta}_{\hat{S}} - \hat{\theta}_{\mathcal{G}} \xrightarrow{p} 0$, whenever the screen retains every cohort with probability approaching one, which by Proposition 4 occurs precisely when no cohort is detectable, $\max_{e<0} |\beta_{g,\mathrm{pre}}(e)| \le c$ for every g; when some cohorts are detectable and dropped, the estimators converge to a common value if and only if the limiting selected set S^* satisfies the equal-targets condition above.*

Proof. Both θ_S and $\theta_{\mathcal{G}}$ are weighted averages of the same cohort effects $\{\theta_g\}$, so they agree exactly when the displayed identity holds; $S = \mathcal{G}$ makes the two averages coincide term by term, and $\theta_g \equiv \theta$ makes each equal θ. Under data-driven selection, if no cohort is detectable then by Proposition 4 the screen retains all of $\mathcal{G}$ with probability approaching one, so $\hat{S} = \mathcal{G}$ on an event of probability tending to one and the two estimators coincide there. In general $\hat{\theta}_{\hat{S}} \xrightarrow{p} \theta_{S^*}$ and $\hat{\theta}_{\mathcal{G}} \xrightarrow{p} \theta_{\mathcal{G}}$, whose difference vanishes exactly under the equal-targets identity applied to S^*. □

Two remarks place the result. First, the equal-targets condition is the precise sense in which the LATT is a distinct estimand: it is distinct exactly to the extent that selection is informative about effect size, and the gap $\theta_S - \theta_{\mathcal{G}}$ is the heterogeneity that Proposition 7 isolates as its composition-gap term Γ, vanishing under homogeneity. This is the equivalence familiar from the local average treatment effect and overlap-weighted estimands, whose subpopulation targets coincide with the population one under an analogous ignorability of selection. Second, target equivalence and estimator convergence come apart. Heterogeneous effects can still leave the estimators convergent, when no cohort is detectable and the entire population is retained, even though the targets would differ for any strict selection; and conversely a strict but ignorable selection can leave the targets equal while the finite-sample estimators differ by the efficiency cost recorded in Proposition 2.

Under data-driven selection the rule (4) couples S to the estimation errors: a confounded cohort whose pre-trend noise happens to fall below c is admitted, and its differential trend enters $\hat{\theta}_S$. This is a pre-test selection bias in the sense of Roth (2022), with a definite source and a definite fate.

Proposition 4 (Pre-test bias)**.** *Under data-driven selection, $\hat{\theta}_S$ carries a finite-sample bias equal to the weighted average of the admitted confounded cohorts' post-treatment differential trends, times their probability of admission. Whether it vanishes turns on detectability. Call a confounded cohort detectable if its population pre-trend statistic exceeds the threshold, $\max_{e<0} |\beta_{g,\mathrm{pre}}(e)| > c$, and* *undetectable if it is flat in the pre-period,* $\max_{e<0}|\beta_{g,\text{pre}}(e)| \leq c$*, yet violates parallel trends in the post-period. As per-cohort information grows, a detectable cohort's admission probability tends to zero and its contribution to the bias vanishes, whereas an undetectable cohort's admission probability tends to one and its contribution persists at every sample size. Hence* $\hat{\theta}_S$ *is consistent for* θ_S *if and only if every confounded cohort is detectable, and otherwise retains an asymptotic bias equal to the residual violation of the undetectable cohorts, which is the identification gap that the honest inference of the next subsection bounds.*

2.4 Honest inference on the selected set

A flat pre-trend does not guarantee a flat post-trend: selection removes gross violations but not subtle residual ones. We therefore pair the point estimate with the sensitivity analysis of Rambachan and Roth (2023), applied to the selected-cohort aggregate. It imposes a restriction $\delta_{S,\text{post}} \in \Delta$ on the residual differential trend and reports the implied confidence set for θ_S. To match the flatness screen we use the level bound $\Delta^{\text{Level}}(M) = \{\delta : |\delta_{\text{post}}(e)| \leq M \ \forall e\}$, which asks how far the treated-comparison difference could have drifted from its flat pre-treatment level. The estimator is then the reweighted post-treatment average, with no extrapolation, and its fixed-length confidence interval (FLCI) is near-optimal (Armstrong and Kolesár, 2018). Because M is in the outcome's own units, the breakdown value M^* is the size of post-treatment violation that would just overturn the conclusion. We benchmark it two ways: against the retained cohorts' own pre-trends, which the screen holds below c in the same units, and against the effect at stake. An M^* several times the retained cohorts' visible pre-trends signals a robust conclusion, and Section 4 reports both benchmarks.

Validity depends on the selection regime. Under ex-ante selection S is independent of $\hat{\beta}$, and the uniform validity of Rambachan and Roth (2023) applies verbatim. Under data-driven selection S is random, and two issues arise. The first is post-selection inference; it is asymptotically harmless when cohorts are well separated from the threshold.

Proposition 5 (Selection consistency and oracle coverage)**.** *Suppose the statistic on which selection is based is consistent for its population counterpart, that the following separation condition holds, that there exists* $\epsilon > 0$ *such that, for every cohort, the population value of that statistic lies at distance at least* ϵ *from the threshold* c*, and that the population credible set's residual violation satisfies the maintained restriction* $\delta_{S^*,\text{post}} \in \Delta^{\text{Level}}(M)$*. Then* $P(\hat{S} = S^*) \to 1$*, where* S^* *denotes the population credible set, and the FLCI computed on the estimated set* $\hat{S}$ *has the same asymptotic coverage as the oracle interval computed on* S^**. Selection consistency delivers this equivalence to the oracle interval, and the maintained restriction delivers the oracle interval's nominal coverage of the causal target, so the two together give nominal coverage.*

Proof. By consistency of the selection statistic together with the separation condition, the probability that any given cohort is misclassified tends to zero. Since the number of cohorts is finite, a union bound gives $P(\hat{S} \neq S^*) \to 0$. On the complementary event the estimator, its covariance estimate and the resulting interval coincide exactly with their oracle counterparts computed on the non-random set S^*. The two sequences therefore share a limiting distribution, and the coverage of the interval computed on $\hat{S}$ converges to that of the oracle interval, which is nominal by the validity of Rambachan and Roth (2023) applied to a fixed set whose residual violation lies in $\Delta^{\text{Level}}(M)$. $\square$

Two caveats attach to Proposition 5. First, the separation condition is untestable and nearly restates the parallel-trends question: to assert that every cohort is plainly credible or plainly not is close to asserting that one already knows which cohorts are credible, so the condition relocates the identifying content rather than supplying it. Second, the guarantee is pointwise, not uniform. Leeb and Pötscher (2005) show that a post-selection estimator's distribution cannot be estimated uniformly consistently, so naive post-selection intervals lack uniform validity (Leeb and Pötscher, 2008). The obstruction is confined to local-to-threshold configurations, where the selection statistic drifts toward the boundary at the sampling rate; away from the boundary, selection is asymptotically deterministic and inference is conventional. A researcher unwilling to assume separation has two options. One secures validity through independence, selecting on pre-determined covariates or an independent sample split, at the cost of the efficiency the split forgoes. The other conditions on the selection and recovers that efficiency. We develop it now, because it delivers the uniform guarantee Proposition 5 lacks.

The construction randomizes the screen and conditions on its outcome. In the reduced-form model $\hat{\beta} \sim \mathcal{N}(\beta, \Sigma)$ with Σ known, draw independent noise $\omega \sim \mathcal{N}(0, \gamma\,\Sigma_{\text{pre}})$ for a scale $\gamma \geq 0$, and select on the noised pre-trends, $\hat{S} = \{g : \max_e |\hat{\beta}_{g,\text{pre}}(e) + \omega_g(e)| \leq c\}$, while estimating the aggregate $\theta_S = \ell_S' \beta_{\text{post}}$ from the unrandomized $\hat{\theta}_S = \ell_S' \hat{\beta}_{\text{post}}$, where ℓ_S carries the reweighting onto S. The target is the reduced-form aggregate: under parallel trends for S it is the causal θ_S, and otherwise it is that quantity plus the residual violation the level bound handles below. Selection uncertainty and identification are thus carried by separate instruments.

Proposition 6 (Uniformly valid carved inference)**.** *Fix $\gamma \geq 0$. Condition on the selection event $\{\hat{S} = S\}$ together with the active coordinates of the flatness screen and their signs—for each retained cohort the pre-period constraints that bind, and for each dropped cohort a pre-period e_g^* whose coordinate breaches the threshold together with its sign—which renders the event a polyhedron in the Gaussian vector even with several pre-periods per cohort. Then $\hat{\theta}_S$ has a truncated Gaussian law,*

$$\hat{\theta}_S \mid \{\hat{S} = S\} \ \sim\ \mathcal{TN}\big(\theta_S,\ s_S^2,\ [\mathcal{V}^-, \mathcal{V}^+]\big), \qquad s_S^2 = \ell_S' \Sigma_{\text{post}}\, \ell_S,$$

whose truncation limits $[\mathcal{V}^-, \mathcal{V}^+]$ are computable in closed form from Σ, γ, and the realized data by the polyhedral lemma of Lee et al. (2016): they are the intersection, over retained cohorts of the two-sided constraints $|\hat{\beta}_{g,\text{pre}} + \omega_g| \leq c$ and over dropped cohorts of the one-sided active constraint $\pm(\hat{\beta}_{g,\text{pre}}(e_g^*) + \omega_g) > c$, of the intervals these place on $\hat{\theta}_S$ once the component of each constraint orthogonal to $\hat{\theta}_S$ is held fixed; with a single pre-period per cohort the retained-cohort constraints alone bind. Inverting the truncated-normal pivot $F^{[\mathcal{V}^-,\mathcal{V}^+]}_{\theta_S, s_S^2}(\hat{\theta}_S)$ yields an interval $\mathcal{C}_{1-\alpha}(S)$ with exact conditional coverage $\Pr(\theta_S \in \mathcal{C}_{1-\alpha}(S) \mid \hat{S} = S) = 1 - \alpha$ for every S and every β, hence exact *unconditional coverage* $\Pr(\theta_S \in \mathcal{C}_{1-\alpha}(\hat{S})) = 1-\alpha$*, uniformly in* β *and without any separation condition. The scale* γ *is a power parameter, not a validity parameter:* $\gamma = 0$ *gives the fully conditional interval,* $\gamma \to \infty$ *drives selection independent of* $\hat{\theta}_S$ *so that the naive interval becomes valid, and intermediate* γ *carves, recovering efficiency that full conditioning spends while retaining exact validity.*

Proof. For a retained cohort the screen $\max_e |\hat{\beta}_{g,\text{pre}}(e) + \omega_g| \le c$ is a box, an intersection of half-spaces. For a dropped cohort the complementary event $\max_e |\cdot| > c$ is a *union* of half-spaces and is not itself convex; conditioning on a breaching coordinate e_g^* and its sign selects the single half-space $\pm(\hat{\beta}_{g,\text{pre}}(e_g^*) + \omega_g) > c$, and when several coordinates breach we condition on the full active set. The conditioned selection event is thus an intersection of half-spaces, a polyhedron in $(\hat{\beta}, \omega)$, for any number of pre-periods per cohort. This conditioning only refines the event, so the conditional pivot stays exact and unconditional coverage is recovered by averaging over the finer partition as well as over S. The pair ($\hat{\theta}_S$, selection) is jointly Gaussian, and decomposing the constraint vector into its component along $\hat{\theta}_S$ and an orthogonal remainder that is independent of $\hat{\theta}_S$, the event fixes the remainder and confines $\hat{\theta}_S$ to an interval $[\mathcal{V}^-, \mathcal{V}^+]$ whose endpoints are the tightest of the resulting linear bounds; this is the polyhedral lemma of Lee et al. (2016). Hence the conditional law of $\hat{\theta}_S$ is $\mathcal{N}(\theta_S, s_S^2)$ truncated to that interval, the pivot is uniform on $[0,1]$ conditional on $\hat{S} = S$, and inverting it gives an interval with conditional coverage $1-\alpha$ for every S and every β. Averaging over S gives the same unconditional coverage, and since the finite-sample pivot is exact for known Σ regardless of where β sits relative to the threshold, coverage is uniform. The endpoint claims in γ hold because $\gamma = 0$ leaves the selection a deterministic function of $\hat{\beta}_{\text{pre}}$, while as $\gamma \to \infty$ the correlation between the selection variable and $\hat{\theta}_S$ vanishes, so $[\mathcal{V}^-, \mathcal{V}^+]$ expands to the real line and the truncated law returns to $\mathcal{N}(\theta_S, s_S^2)$. □

Proposition 6 upgrades the pointwise oracle coverage of Proposition 5 to a uniform guarantee, precisely across the local-to-threshold configurations that obstruct it. It does so for the sampling and selection uncertainty in the reduced-form aggregate; the identification gap remains with the level bound, which we now fold in to reach the causal target.

Corollary 1 (Honest causal interval under data-driven selection)**.** *Let* $\theta^{\text{c}}_{\hat{S}} = \ell'_{\hat{S}} \tau_{\text{post}}$ *be the causal target on the selected set and* $B_{\hat{S}} = \sup_{\delta_{\text{post}} \in \Delta^{\text{Level}}(M)} |\ell'_{\hat{S}} \delta_{\text{post}}|$ *the worst-case residual bias the level bound allows, so that* $|\theta_{\hat{S}} - \theta^{\text{c}}_{\hat{S}}| \le B_{\hat{S}}$ *whenever* $\delta_{\hat{S},\text{post}} \in \Delta^{\text{Level}}(M)$*; for a single-period or convex-averaging functional* $a(\cdot)$*,* $B_{\hat{S}} = M$*. Then the widened carved interval*

$$\mathcal{C}^M_{1-\alpha}(\hat{S}) \;:=\; \big[\, \inf \mathcal{C}_{1-\alpha}(\hat{S}) - B_{\hat{S}},\; \sup \mathcal{C}_{1-\alpha}(\hat{S}) + B_{\hat{S}} \,\big]$$

covers the causal target, $\Pr\big(\theta^{\text{c}}_{\hat{S}} \in \mathcal{C}^M_{1-\alpha}(\hat{S})\big) \ge 1-\alpha$*, uniformly in* β *and without a separation condition, whenever* $\delta_{\hat{S},\text{post}} \in \Delta^{\text{Level}}(M)$*.*

Proof. Proposition 6 gives $\theta_{\hat{S}} \in \mathcal{C}_{1-\alpha}(\hat{S})$ with probability $1-\alpha$. The maintained restriction places $\theta^{\text{c}}_{\hat{S}}$ within $B_{\hat{S}}$ of $\theta_{\hat{S}}$ deterministically, so $\{\theta_{\hat{S}} \in \mathcal{C}_{1-\alpha}(\hat{S})\} \subseteq \{\theta^{\text{c}}_{\hat{S}} \in \mathcal{C}^M_{1-\alpha}(\hat{S})\}$, and monotonicity of probability gives the bound. $\square$

The guarantee is one-sided by design: $\mathcal{C}^M_{1-\alpha}(\hat{S})$ is *conservative*, superposing the exact carved interval for the reduced-form $\theta_{\hat{S}}$ on the worst-case additive bias $B_{\hat{S}}$ rather than folding sampling, selection, and identification uncertainty into a single length-optimal object. It is accordingly *not* the near-optimal fixed-length interval of Armstrong and Kolesár (2018); that construction and its optimality belong to the fixed-set case of Theorem 1, where selection does not randomize the target and the bias enters only through the standard error. Under data-driven selection we accept a modest excess length in exchange for a single interval valid against both the selection and the residual violation. Simulations confirm this. A scalar illustration—one retained cohort with a single pre-period, so the estimate and the selection statistic are jointly bivariate normal and the mechanism is transparent—makes the point: at a pre–post sampling correlation of 0.7 the naive post-selection interval covers at 0.92 and then 0.78 as the cohort's population pre-trend moves one and two thresholds off center, while the carved interval holds between 0.93 and 0.96 throughout. The same construction at four pre-periods—using the box screen and the violating-coordinate conditioning of the proof—likewise restores near-nominal coverage where the naive post-selection interval undercovers by several points, confirming that the guarantee is not special to the scalar case. The multi-cohort, four-pre-period design of Section 3.5 exhibits the same understatement of naive uncertainty under estimated selection. The randomization also recovers power a split would forfeit: at the worst configuration the carved interval at $\gamma = 0.5$ is about a quarter narrower than the fully conditional one at $\gamma = 0$, at unchanged coverage.

The second issue is that the bound M must accommodate a random selected set. The FLCI at M is valid only if the selected set's residual violation lies in $\Delta^{\text{Level}}(M)$. Since that violation is itself random, unconditional coverage requires M to dominate it with high probability.

Lemma 1 (Calibration to the random selected set)**.** *Under data-driven selection the selected aggregate's residual violation is random, and the FLCI's unconditional coverage is governed by the joint distribution of that violation, the selection event, and the sampling error. When the violation's dispersion is non-negligible relative to the FLCI's sampling slack, setting M to the mean residual violation undercovers, because the realized violation exceeds M in a nontrivial fraction of samples, and restoring nominal unconditional coverage requires M to exceed the mean and approach an upper quantile of the residual-violation distribution, with the sampling slack providing a partial buffer. When the slack instead dominates the dispersion the buffer alone can suffice.*

The practical implication, in the regime where the violation's spread is the binding consideration, is that M should be calibrated to the residual violation of a typical selected set, not to its mean.

Finally, we caution against one otherwise-natural choice of Δ. The relative-magnitudes restriction $\Delta^{RM}(\bar{M})$ of Rambachan and Roth (2023), following Manski and Pepper (2018), bounds post-treatment violations by $\bar{M}$ times the observed maximal pre-treatment violation, so its identified-set half-width is proportional to the observed pre-trend. When pre-trends are uninformative, they are small while the true violation need not be, so the interval width tends to zero even as the bias does not—undercoverage precisely where our method's value is in question. The fixed bound $\Delta^{\text{Level}}(M)$, set by the researcher rather than by the observed pre-trends, avoids this.

Remark 1 (Estimated covariance). Proposition 6 and the half-width (13) treat Σ as known; in practice one plugs in a consistent estimate $\hat{\Sigma}$, which enters twice, in the aggregate standard error s_S and in the selection geometry $[\mathcal{V}^-, \mathcal{V}^+]$ and carving scale, so its dependence is more delicate here than in a fixed-target sensitivity analysis, where Σ enters only the standard error. Suppose $\hat{\Sigma} \xrightarrow{p} \Sigma$ and the separation condition of Proposition 5 holds, so that each cohort's population pre-trend statistic lies a fixed distance from the threshold. The truncated-normal pivot $F^{[\mathcal{V}^-,\mathcal{V}^+]}_{\theta_S, s_S^2}(\hat{\theta}_S)$ is continuous in $(\Sigma, \mathcal{V}^-, \mathcal{V}^+)$ at the truth, its truncation limits and scale are continuous functions of $\hat{\Sigma}$, and under separation the selection event is correctly classified with probability approaching one; by Slutsky the pivot evaluated at $\hat{\Sigma}$ converges in law to Uniform$[0, 1]$, so the carved interval has asymptotic coverage $1 - \alpha$. The known-Σ exactness of Proposition 6 thus degrades only to first-order valid coverage under estimation, and the panel simulation of Section 3.5, which estimates the covariance from the data rather than supplying it, bears this out. The separation condition is not cosmetic: within a $\hat{\Sigma}$-sampling neighbourhood of the threshold a cohort's inclusion is itself estimated, and plug-in post-selection coverage there carries a higher-order, non-uniform cost, the estimated-Σ shadow of the Leeb–Pötscher obstruction. A researcher unwilling to assume separation should accordingly hold the carving scale γ away from its degenerate limit, or use the independent-split construction, either of which makes the selection insensitive to $\hat{\Sigma}$ at the boundary.

2.5 When the trade pays: a dominance condition

The scope condition seen in the simulations has a closed-form decision-theoretic counterpart. Changing the estimand is worth doing only when the bias it removes outweighs the variance and the change of target it costs. Three forces set the sign: the violation the screen removes, the variance that dropping a cohort costs, and the heterogeneity between the retained subpopulation and the full one. The sign turns on their balance—and, crucially, not on the informativeness of the pre-trends, which governs only how much of a fixed-sign advantage is realized.

We state the comparison for K cohorts, and then read from the two-cohort special case the one thing the general result leaves implicit—how the advantage scales with the informativeness of the pre-trends. Let each cohort g carry a weight w_g (normalized so $\sum_g w_g = 1$), a causal effect $\theta_g = \text{ATT}(g, \cdot)$, a post-treatment violation V_g (the aggregate $a(\delta_{g,\text{post}})$, zero for a clean cohort), and an independent sampling variance σ_g^2 for its aggregated post-treatment estimate. The screen retains cohort g with probability π_g, independently across cohorts and of the post-treatment estimates. The ATT estimator is the fixed average $\sum_g w_g \hat{\beta}_{g,\text{post}}$; the LATT estimator reweights over the retained set $\hat{S}$, $\sum_{g \in \hat{S}} w_g \hat{\beta}_{g,\text{post}} / W_{\hat{S}}$ with $W_S = \sum_{g \in S} w_g$.

Proposition 7 (General-K dominance)**.** *With the target $\theta_{\mathcal{G}} = \sum_g w_g \theta_g$, the ATT estimator has* *mean squared error* $\mathrm{MSE}_{\mathrm{ATT}} = \big(\sum_g w_g V_g\big)^2 + \sum_g w_g^2 \sigma_g^2$, *and the LATT estimator has*

$$\mathrm{MSE}_{\mathrm{LATT}} = \sum_{S \subseteq \mathcal{G}} \Big(\prod_{g\in S} \pi_g \prod_{g\notin S}(1-\pi_g)\Big)\Big[\big(\theta_S - \theta_{\mathcal{G}} + B_S\big)^2 + \mathrm{Var}_S\Big], \tag{5}$$

where $\theta_S = \sum_{g\in S} w_g \theta_g / W_S$, $B_S = \sum_{g\in S} w_g V_g / W_S$, *and* $\mathrm{Var}_S = \sum_{g\in S} w_g^2 \sigma_g^2 / W_S^2$, *with the convention* $S = \mathcal{G}$ *when the screen retains no cohort. When every cohort is retained,* $\pi_g = 1$, *the two estimators coincide and* $\Delta = \mathrm{MSE}_{\mathrm{ATT}} - \mathrm{MSE}_{\mathrm{LATT}} = 0$. *Suppose instead the screen separates the cohorts, discarding every confounded cohort,* $\pi_g \to 0$ *for* $V_g \neq 0$, *and retaining every clean cohort,* $\pi_g \to 1$ *for* $V_g = 0$, *so that the retained set converges in probability to the clean set* $\mathcal{C}$ *and the dropped set is the confounded set* $\mathcal{F} = \mathcal{G} \setminus \mathcal{C}$. *Then*

$$\Delta \;\longrightarrow\; \Delta^* \;=\; B_{\mathcal{F}}^2 - \Gamma^2 - \Delta\mathrm{Var}, \tag{6}$$

where $B_{\mathcal{F}} = \sum_{g\in\mathcal{F}} w_g V_g$ *is the aggregate violation removed by selection,* $\Gamma = \theta_{\mathcal{C}} - \theta_{\mathcal{G}}$ *is the composition gap between the retained subpopulation and the full population, and* $\Delta\mathrm{Var} = \mathrm{Var}_{\mathcal{C}} - \sum_g w_g^2\sigma_g^2 \geq 0$ *is the variance cost of dropping. Hence in the informative limit the LATT dominates the ATT in mean squared error if and only if*

$$B_{\mathcal{F}}^2 \;>\; \Gamma^2 + \Delta\mathrm{Var}. \tag{7}$$

With two equally weighted cohorts, one clean and one confounded with violation V, *effect gap* γ, *and common variance* σ^2, *one has* $B_{\mathcal{F}}^2 = V^2/4$, $\Gamma^2 = \gamma^2/4$, *and* $\Delta\mathrm{Var} = \sigma^2/2$, *so* (7) *becomes* $V^2 > \gamma^2 + 2\sigma^2$; *Corollary 2 below develops this two-cohort case, where the informativeness of the pre-trends enters.*

Proof. The ATT estimator is a fixed linear combination with mean $\theta_{\mathcal{G}} + \sum_g w_g V_g$ and variance $\sum_g w_g^2\sigma_g^2$, giving $\mathrm{MSE}_{\mathrm{ATT}}$. For the LATT, condition on the retention configuration; on the event that the retained set is S, which has probability $\prod_{g\in S}\pi_g \prod_{g\notin S}(1-\pi_g)$, the estimator is $\sum_{g\in S} w_g \hat{\beta}_{g,\mathrm{post}}/W_S$, whose mean is $\theta_S + B_S$ and whose variance is Var_S, since retention is independent of the post-treatment estimates. Its error relative to $\theta_{\mathcal{G}}$ has squared bias $(\theta_S - \theta_{\mathcal{G}} + B_S)^2$ and variance Var_S; averaging over configurations gives (5). At $\pi_g = 1$ only $S = \mathcal{G}$ has mass, where $\theta_{\mathcal{G}} - \theta_{\mathcal{G}} + B_{\mathcal{G}} = \sum_g w_g V_g$ and $\mathrm{Var}_{\mathcal{G}} = \sum_g w_g^2\sigma_g^2$, so $\mathrm{MSE}_{\mathrm{LATT}} = \mathrm{MSE}_{\mathrm{ATT}}$ and $\Delta = 0$. Under separation the configuration $S = \mathcal{C}$ carries probability approaching one, so $\mathrm{MSE}_{\mathrm{LATT}} \to (\theta_{\mathcal{C}} - \theta_{\mathcal{G}})^2 + \mathrm{Var}_{\mathcal{C}}$, using $B_{\mathcal{C}} = 0$ because clean cohorts have $V_g = 0$. Since clean cohorts contribute nothing to $\sum_g w_g V_g$, that sum equals $B_{\mathcal{F}}$, and $\Delta \to B_{\mathcal{F}}^2 + \sum_g w_g^2\sigma_g^2 - \Gamma^2 - \mathrm{Var}_{\mathcal{C}} = B_{\mathcal{F}}^2 - \Gamma^2 - \Delta\mathrm{Var}$, which is (6); the threshold (7) is its positivity, and the two-cohort reduction is the displayed substitution. □

The general threshold (7) is the two-cohort comparison written for many cohorts, each of its three forces now an aggregate. The reason to trade is the total violation $B_{\mathcal{F}}$ selection removes, not any single cohort's. The estimand cost is the composition gap Γ between the retained subpopulation's effect and the full population's, the many-cohort form of γ; it vanishes under the equal-targets condition of Proposition 3. The variance cost $\Delta\mathrm{Var}$ is the extra sampling variance from reweighting onto fewer cohorts. So the two-cohort reading holds at any cohort count: the sign of the advantage is a violation-against-noise-and-heterogeneity question, settled by (7) and independent of the informativeness of the pre-trends, which governs only magnitude—Δ runs from zero when the screen retains everyone to Δ^* when it separates the cohorts. Two qualifications attach and connect to the inference above. The separation that makes (6) exact is the mean-squared-error counterpart of the condition behind Proposition 5; near-threshold cohorts, whose population pre-trend sits within sampling error of c, are retained with interior probability and are the same local-to-threshold configurations that obstruct uniform post-selection inference. And a confounded cohort that is undetectable, flat in the pre-period yet violating parallel trends after, is retained in the limit rather than dropped, so its violation stays in both estimators and in the residual bound B_{S^*}; it enters (6) by replacing $B_{\mathcal{F}}$ with the detectable cohorts' violation and Γ with Γ augmented by that residual, which is the identification gap of Proposition 4 that the honest inference of Section 2.4 bounds.

The two-cohort case makes the three forces explicit and, uniquely, traces the advantage along the informativeness axis. Consider two equally weighted cohorts sharing a common sampling variance σ^2, observed as one pre- and one post-treatment coefficient each, with independent pre- and post-period sampling errors. One cohort is clean, $\delta_1 = 0$; the other is confounded, with a post-treatment violation $V > 0$ and a pre-treatment violation ϕV, where $\phi \geq 0$ is the informativeness of the pre-trend. Their treatment effects are τ_1 and τ_2, with $\gamma := \tau_2 - \tau_1$, so the target is the average $\bar{\tau} = (\tau_1 + \tau_2)/2$. Taking c large enough that the clean cohort is retained with probability approaching one, the confounded cohort survives the screen $|\hat{\beta}_{2,\text{pre}}| \leq c$ with probability

$$p_2(\phi) \;=\; \Phi\Big(\tfrac{c-\phi V}{\sigma}\Big) - \Phi\Big(\tfrac{-c-\phi V}{\sigma}\Big), \tag{8}$$

which decreases from near one, when the confound is invisible in the pre-period, to zero as the pre-trend announces it.

Corollary 2 (Informativeness scaling, two cohorts)**.** *In the two-cohort model, with the clean cohort retained with probability approaching one, the mean-squared-error advantage of the LATT estimator over the ATT estimator is*

$$\Delta(\phi) \;=\; \mathrm{MSE}_{\mathrm{ATT}} - \mathrm{MSE}_{\mathrm{LATT}} \;=\; \frac{1-p_2(\phi)}{4}\,\big[\,V^2 - \gamma^2 - 2\sigma^2\,\big], \tag{9}$$

the general advantage (6) *specialized by* $B_{\mathcal{F}}^2 = V^2/4$, $\Gamma^2 = \gamma^2/4$, $\Delta\mathrm{Var} = \sigma^2/2$ *and scaled by the survival factor* $1 - p_2(\phi)$*. Hence the LATT dominates the ATT in mean squared error if and only if*

$$V^2 \;>\; \gamma^2 + 2\sigma^2. \tag{10}$$

The sign is fixed by (10) *and is independent of the informativeness* ϕ*;* ϕ *enters only through* $1 - p_2(\phi)$*, nondecreasing in* ϕ*, which scales a fixed-sign advantage from zero, when the pre-trend is uninformative, to its ceiling* $\frac{1}{4}[V^2 - \gamma^2 - 2\sigma^2]$ *as* $\phi \to \infty$*.*

Proof. Write $\tau_1 = \bar{\tau} - \gamma/2$, $\tau_2 = \bar{\tau} + \gamma/2$, so $\hat{\beta}_{1,\text{post}} \sim \mathcal{N}(\bar{\tau} - \gamma/2, \sigma^2)$ and $\hat{\beta}_{2,\text{post}} \sim \mathcal{N}(\bar{\tau} + \gamma/2 + V, \sigma^2)$, independent of each other and of the retention indicators. The ATT estimator has $\mathrm{MSE}_{\mathrm{ATT}} = V^2/4+\sigma^2/2$. With the clean cohort always retained, the LATT estimator is the two-cohort average with probability p_2 and $\hat{\beta}_{1,\mathrm{post}}$ with probability $1-p_2$, so $\mathrm{MSE}_{\mathrm{LATT}} = p_2(V^2/4+\sigma^2/2)+(1-p_2)(\gamma^2/4+\sigma^2)$; subtracting gives (9), which is (6) under the stated substitution. Since $1 - p_2 \geq 0$ the sign is that of the bracket, giving (10), and $p_2(\phi)$ is nonincreasing in ϕ for $\phi V \geq 0$, so $1 - p_2$ is nondecreasing. $\square$

The bracket's three terms are the forces of (7) in closed form: $+V^2$ the bias the screen removes, $-2\sigma^2$ the variance that dropping a cohort costs, and $-\gamma^2$ the estimand-change cost. As set out when the estimand was defined (Section 2), the LATT is our target, against which the estimator is unbiased; the comparison adopts the ATT as the common target as a courtesy to a reader who insists on it, and $-\gamma^2$ is what that reader is charged. It vanishes under homogeneity, $\gamma = 0$, the normalization used in the simulations, the most favorable case for the method. Two readings follow. First, informative pre-trends are necessary but not sufficient: if $V^2 < \gamma^2 + 2\sigma^2$, no degree of informativeness rescues the LATT, since $1 - p_2$ only drives an already-negative bracket further from zero—informativeness scales the advantage, it does not create one. Second, when the violation is large enough to make the bracket positive, the advantage rises monotonically in informativeness, from zero at $\phi = 0$ to the bracket's value, the vertical gap between the two bias curves in Figure 1.

The dominance condition can be read off the data, up to the one quantity it cannot identify, and so furnishes a diagnostic for whether the trade is worth making. The variance cost $\Delta\mathrm{Var}$ is a function of Σ and the weights alone and is therefore estimable, and the difference between the two point estimators consistently estimates the violation net of the composition gap,

$$\hat{\theta}_{\mathcal{G}} - \hat{\theta}_{\hat{S}} \xrightarrow{p} B_{\mathcal{F}} - \Gamma =: D, \tag{11}$$

since the ATT estimator carries the dropped cohorts' violation while the LATT estimator does not. What is not identified is Γ on its own, because it depends on the counterfactual effects of the very cohorts the screen discards. Writing $B_{\mathcal{F}} = D + \Gamma$ in (7), the advantage $\Delta^* = B_{\mathcal{F}}^2 - \Gamma^2 - \Delta\mathrm{Var}$ becomes $D^2 + 2D\Gamma - \Delta\mathrm{Var}$, linear in the unidentified Γ because the squared composition gap cancels. Two readings follow. Under homogeneous effects, $\Gamma = 0$, the trade pays exactly when the squared gap between the estimators exceeds the variance cost,

$$D^2 > \Delta\mathrm{Var}, \tag{12}$$

a comparison the practitioner can compute and, using the carved standard error of Proposition 6 for D, test. In general the verdict is monotone in Γ, increasing when $D > 0$, and flips at the breakdown value $\Gamma^* = (\Delta\mathrm{Var} - D^2)/(2D)$. A researcher thus reports $\hat{D}$, $\widehat{\Delta\mathrm{Var}}$, and Γ^*, and the trade pays unless the dropped cohorts' effects differ from the retained ones by more than Γ^* allows—a sensitivity statement in the same idiom as the breakdown value M^* for the identifying restriction, now for the estimand change rather than the violation.

These pieces assemble into a rule the practitioner can run: report *both* honest intervals—for the ATT and for the LATT—with the diagnostic, and let the rule advise which is more informative.

Because each interval covers its own target whatever the data say, ranking them can misfire on efficiency but never on coverage—feasible up to the single quantity, the effects of the discarded cohorts, that no design can identify.

Proposition 8 (A feasible and honest switching rule)**.** *Consider the informative limit of Proposition 7, and suppose the researcher reports the honest level-bound interval for* both *the ATT and the LATT, together with the rule: prefer the LATT when* $\hat{D}^2 > \widehat{\Delta\mathrm{Var}}$ *and the ATT otherwise, where* $\hat{D} = \hat{\theta}_{\mathcal{G}} - \hat{\theta}_{\hat{S}} \xrightarrow{p} D = B_{\mathcal{F}} - \Gamma$ *and* $\widehat{\Delta\mathrm{Var}}$ *is the variance cost, a function of* Σ *and the weights alone. Then:*

(i) (Honesty is unconditional.) *Each reported interval covers its own target at level* $1-\alpha$ *for every* β*, by the validity of the level-bound interval (carved, under data-driven selection, by Proposition 6). Since both are always reported, the rule—however it decides—cannot affect coverage; it ranks the two honest intervals and is an efficiency device only.*

(ii) (Consistency under homogeneity.) *If treatment effects are homogeneous, so the composition gap* $\Gamma = 0$*, and* $D^2 \neq \Delta\mathrm{Var}$*, the rule prefers the mean-squared-error-dominant procedure with probability approaching one.*

(iii) (Sensitivity under heterogeneity.) *In general the advantage* $\Delta^* = B_{\mathcal{F}}^2 - \Gamma^2 - \Delta\mathrm{Var} = D^2 + 2D\Gamma - \Delta\mathrm{Var}$ *is monotone in the unidentified* Γ *when* $D > 0$ *and changes sign at the breakdown value* $\Gamma^* = (\Delta\mathrm{Var} - D^2)/(2D)$*. The trade pays unless the discarded cohorts' effects differ from the retained cohorts' by more than* Γ^**.*

Proof. (i) Each interval's coverage statement is a property of that interval alone and holds for every β; reporting both leaves each untouched, so whichever the rule recommends, the reader is handed an interval that covers at $1-\alpha$. (ii) By Proposition 7, in the informative limit the LATT dominates the ATT in mean squared error iff $\Delta^* > 0$, and under $\Gamma = 0$ this is $D^2 > \Delta\mathrm{Var}$. Since $\hat{D} \xrightarrow{p} D$ and $\widehat{\Delta\mathrm{Var}} \xrightarrow{p} \Delta\mathrm{Var}$ (indeed the latter is known given Σ), the continuous map gives $\mathbf{1}\{\hat{D}^2 > \widehat{\Delta\mathrm{Var}}\} \xrightarrow{p} \mathbf{1}\{D^2 > \Delta\mathrm{Var}\}$ whenever $D^2 \neq \Delta\mathrm{Var}$, which is the sign of Δ^*. (iii) Writing $B_{\mathcal{F}} = D + \Gamma$ in $\Delta^* = B_{\mathcal{F}}^2 - \Gamma^2 - \Delta\mathrm{Var}$ cancels the squared gap and leaves $\Delta^* = D^2 + 2D\Gamma - \Delta\mathrm{Var}$, linear in Γ with slope $2D$; its root is Γ^*, and it is increasing in Γ when $D > 0$. $\square$

Proposition 8 turns the informal "report the LATT when pre-trends are informative" into an estimable statistic, and separates what the data decide from what they cannot: D and the variance cost are estimable, so under homogeneity the decision is made for the researcher, while the unidentified composition gap Γ is reported as the breakdown Γ^*, read as M^* is for the identifying restriction. The rule never suppresses an interval, so it cannot cost coverage.

Three caveats keep the dominance calculation honest. The advantage (9) is an upper bound on what the method delivers, because it omits two channels that both work against the LATT: occasional false rejection of the clean cohort, which enters at order $1 - p_1$ when the threshold is not comfortably large, and the additional variance that data-driven selection induces and that a standard error conditional on the realized set does not capture, the understatement documented in Section 3.5. Both shrink the realized advantage without changing its sign or its shape. Finally, (9) is a comparison of point estimators under squared-error loss, not of the honest intervals that are the paper's recommended output. The interval analogue replaces the reason-to-trade V with the residual violation net of the calibration buffer of Lemma 1, and the variance cost σ^2 with a sampling slack inflated by the selection quantile, so that the level-bound half-width comparison turns on the same three forces with V discounted and σ inflated. The mean-squared-error threshold (10) correctly signs that comparison and is the transparent summary; the interval version, to which we now turn, sharpens the constants.

The comparison of honest intervals can be made exact in the same two-cohort model, in the informative regime where the screen discards the confounded cohort with probability approaching one, so that the LATT selects the clean cohort alone. This is the regime in which the point-estimate advantage of Corollary 2 attains its ceiling, and it is where the method is meant to help. Both procedures report a level-bound fixed-length confidence interval, whose half-width for a scalar target estimated with standard error s under a worst-case bias bound B is

$$h(B, s) \;=\; s\,\mathrm{cv}_\alpha\big(B/s\big), \qquad \mathrm{cv}_\alpha(t) = \text{the } (1-\alpha) \text{ quantile of } |\mathcal{N}(t, 1)|, \tag{13}$$

the near-optimal construction of Armstrong and Kolesár (2018); $\mathrm{cv}_\alpha(0) = z_{1-\alpha/2}$ is the pure-noise critical value and $\mathrm{cv}_\alpha(t) \uparrow t + z_{1-\alpha}$ as the bias comes to dominate. The LATT, using the clean cohort alone, is unbiased with standard error σ and a residual-violation bound M_L, giving half-width $\sigma\,\mathrm{cv}_\alpha(M_L/\sigma)$. The ATT estimator, using both cohorts, has standard error $\sigma/\sqrt{2}$ but must set its bound to cover its own aggregate violation, $M_A = V/2$, giving half-width $(\sigma/\sqrt{2})\,\mathrm{cv}_\alpha\big(V/(\sqrt{2}\,\sigma)\big)$. Each interval covers the common target at its nominal level by construction, since each bound dominates its own aggregate's violation, so the honest comparison is not one of coverage but of expected length at equal coverage.

Before specializing to two cohorts, we state the comparison in general, since it holds at any cohort count and isolates the single quantity—the violation carried by the dropped cohorts—that decides it. Write the ATT estimator as $\hat{\theta}_{\mathcal{G}} = \ell_{\mathcal{G}}'\hat{\beta}_{\text{post}}$ with aggregate standard error $s_{\mathcal{G}}^2 = \ell_{\mathcal{G}}'\Sigma_{\text{post}}\ell_{\mathcal{G}}$ and honest bias bound $B_{\mathcal{G}}$ dominating the full set's violation $|\ell_{\mathcal{G}}'\delta_{\text{post}}|$, and the LATT estimator as $\hat{\theta}_S = \ell_S'\hat{\beta}_{\text{post}}$ with carved standard error s_S and residual bound B_S. Both honest intervals are the fixed-length construction (13).

Theorem 1 (Width dominance)**.** *Suppose the LATT interval is the carved interval of Proposition 6 (under ex-ante selection, the interval of Rambachan and Roth, 2023), with half-width $h_S = h(B_S, s_S)$, and let $h_{\mathcal{G}} = h(B_{\mathcal{G}}, s_{\mathcal{G}})$ be the ATT interval. Then*

$$h_S < h_{\mathcal{G}} \quad\Longleftrightarrow\quad s_S\,\mathrm{cv}_\alpha\big(B_S/s_S\big) \;<\; s_{\mathcal{G}}\,\mathrm{cv}_\alpha\big(B_{\mathcal{G}}/s_{\mathcal{G}}\big).$$

Holding $(s_{\mathcal{G}}, s_S, B_S)$ fixed, there is a unique threshold

$$B^*_{\mathcal{G}} \;=\; s_{\mathcal{G}}\,\mathrm{cv}_\alpha^{-1}\!\Big(\tfrac{s_S}{s_{\mathcal{G}}}\,\mathrm{cv}_\alpha\big(B_S/s_S\big)\Big)$$

such that the credible-subpopulation interval is strictly shorter than the ATT interval if and only if $B_{\mathcal{G}} > B^*_{\mathcal{G}}$. Since $B_{\mathcal{G}}$ carries the dropped cohorts' aggregate violation $B_{\mathcal{F}}$ while B_S does not, this is a threshold on $B_{\mathcal{F}}$; and whenever restricting to S raises the aggregate variance, $s_S > s_{\mathcal{G}}$, the threshold is strictly positive, so a minimum dropped violation is required before bias relief outweighs the variance cost.

Proof. By (13) each interval is the near-optimal fixed-length interval of Armstrong and Kolesár (2018), with half-width $h(B,s) = s\,\mathrm{cv}_\alpha(B/s)$ and coverage at least $1-\alpha$ whenever the bound dominates the estimator's bias; for the ATT this gives $h_{\mathcal{G}} = h(B_{\mathcal{G}}, s_{\mathcal{G}})$, and under ex-ante S the same applies to the LATT. Under data-driven selection $\hat\theta_S \mid \{\hat S = S\}$ is truncated Gaussian, so the plug-in interval is not honest; Proposition 6 replaces it with the carved statistic, whose pivot is exactly $\mathcal{N}(\theta_S, s_S^2)$-calibrated unconditionally and uniformly in β, so the carved interval is again of the form (13) with standard error s_S and bound B_S, giving $h_S = h(B_S, s_S)$. It is this step that licenses the comparison: without it h_S is not an honest half-width, and it is the step the fixed-target sensitivity analysis never takes. Both half-widths share $\varphi(B,s) = s\,\mathrm{cv}_\alpha(B/s)$, so $h_S < h_{\mathcal{G}}$ is the displayed inequality. Fix $(s_{\mathcal{G}}, s_S, B_S)$ and set $c_0 = s_S\,\mathrm{cv}_\alpha(B_S/s_S)$. The map $g(B) = s_{\mathcal{G}}\,\mathrm{cv}_\alpha(B/s_{\mathcal{G}})$ is continuous and strictly increasing, from $g(0) = s_{\mathcal{G}}\,z_{1-\alpha/2}$ to ∞; since $\mathrm{cv}_\alpha : [0,\infty) \to [z_{1-\alpha/2}, \infty)$ is a strictly increasing bijection and $c_0/s_{\mathcal{G}} \geq (s_S/s_{\mathcal{G}})\,z_{1-\alpha/2} \geq z_{1-\alpha/2}$, the equation $g(B^*_{\mathcal{G}}) = c_0$ has the unique solution displayed, and $h_S < h_{\mathcal{G}} \iff B_{\mathcal{G}} > B^*_{\mathcal{G}}$. Finally $s_S \geq s_{\mathcal{G}}$ because reweighting onto the retained set cannot lower the aggregate variance below the full set's optimum under precision weights, so the inequality $c_0/s_{\mathcal{G}} \geq z_{1-\alpha/2}$ is strict, and $B^*_{\mathcal{G}} > 0$, whenever $s_S > s_{\mathcal{G}}$. □

Corollary 3 (Explicit threshold under equal weights)**.** *Let K equally weighted cohorts with common per-cohort post-treatment variance σ^2 and independent estimates comprise k clean cohorts ($V_g = 0$), retained, and $K-k$ confounded cohorts (violation V), dropped in the informative limit, with the retained set truly flat, $B_S = 0$. Writing $f = (K-k)/K$, so that $s_{\mathcal{G}} = \sigma/\sqrt{K}$, $s_S = \sigma/\sqrt{k}$, and $B_{\mathcal{G}} = fV$,*

$$h_S < h_{\mathcal{G}} \quad\iff\quad \mathrm{cv}_\alpha\!\Big(\tfrac{fV\sqrt{K}}{\sigma}\Big) > \sqrt{\tfrac{K}{k}}\,z_{1-\alpha/2} \quad\iff\quad V > V^* := \frac{\sigma}{f\sqrt{K}}\,\mathrm{cv}_\alpha^{-1}\!\Big(\sqrt{\tfrac{K}{k}}\,z_{1-\alpha/2}\Big),$$

equivalently $B_{\mathcal{F}} = fV > B^*_{\mathcal{F}} = (\sigma/\sqrt{K})\,\mathrm{cv}_\alpha^{-1}\big(\sqrt{K/k}\,z_{1-\alpha/2}\big)$. At $K=2$, $k=1$ this is $V > 1.59\,\sigma$ (at $\alpha = 0.05$).

Two features carry the economics. The width threshold $V > 1.59\,\sigma$ exceeds the mean-squared-error threshold $V > \sqrt{2}\,\sigma \approx 1.41\,\sigma$ of Corollary 2, because the fixed-length interval charges for the standard-error inflation of dropping a cohort, not only for its variance; and coverage is never the margin, since both procedures are honest—the ATT pays for honesty in a width that grows linearly in V, whereas the LATT pays a fixed $2\sigma\, z_{1-\alpha/2}$ set by its variance alone, the price-of-honesty reading of Figure 2. Allowing the retained set a residual bound $M_L > 0$ rather than exact flatness, the LATT interval carries standard error σ and bound M_L against the ATT's $\sigma/\sqrt{2}$ and $M_A = V/2$, so by (13) it is shorter if and only if $\mathrm{cv}_\alpha(M_L/\sigma) < \frac{1}{\sqrt{2}}\,\mathrm{cv}_\alpha(V/(\sqrt{2}\,\sigma))$; a residual $M_L > 0$, and heterogeneity $\gamma \neq 0$ when the target is the ATT (which adds $|\gamma|/2$ to the LATT's effective bias), both raise the threshold, in the direction the point analysis already indicated.

3 Simulation study

3.1 Design

We study the method in two environments. In the reduced-form model we draw event-study coefficients directly from their asymptotic distribution, $\hat{\beta} \sim \mathcal{N}(\tau + \delta, \Sigma)$ with Σ known, which isolates the identification and inference mechanics in the setting where the underlying theory is exact. In the panel model we generate individual outcomes for a staggered design and estimate both the coefficients and their covariance from the data, which confirms that the conclusions survive realistic estimation. Unless stated otherwise, results are averages over eight to twenty thousand replications.

The cohort structure follows a common template. Some cohorts are clean, with $\delta_g = 0$, and the remainder confounded, with a nonzero differential trend, and Table 2 gives the count and clean-confounded split used in each exercise. Every cohort is given the same true treatment effect, normalized to one. This normalization is deliberate, making the true ATT and the true LATT both equal to one, so that any departure of an estimator from one is attributable to a violation of parallel trends rather than to treatment-effect heterogeneity. It therefore isolates the identification problem that the paper is about.

Throughout, each cohort carries a four-period pre-treatment and four-period post-treatment event study with the reference period normalized to zero. A confounded cohort carries a level violation, its treated-comparison difference shifting by $V > 0$ in the post-period and by ϕV in the pre-period, for a scalar $\phi \geq 0$ we call the informativeness parameter and that is the master axis of the study. When $\phi = 0$ the confound is flat before treatment and no pre-trend rule can detect it, and as ϕ grows the pre-period shift announces it and it becomes detectable. Because V is the post-treatment violation it is directly comparable to the level bound M of $\Delta^{\mathrm{Level}}(M)$. In the controlled coverage exercise below the aggregate being tested carries this violation exactly, which makes the boundary statement "covers if and only if $M \geq V$" checkable. More generally, coverage requires only that M dominate the selected aggregate's violation, which for a set mixing clean and confounded cohorts is weakly below the largest single-cohort V, so $M \geq V$ is sufficient and is necessary only in the worst case of an all-confounded set. This also makes ϕ the operational version of the informativeness of pre-trends discussed in Section 2.

The selection rule is the flatness screen, posed in the same currency as the honest inference, so cohort g is retained if its estimated pre-trend is close to flat, $\max_e |\hat{\beta}_{g,\mathrm{pre}}(e)| \leq c$. The estimator of

Table 2: Simulation parameters. Every exercise uses a four-period pre and four-period post event study with reference period zero, equal cohort weights, and true effect one for every cohort. The reduced-form model draws coefficients directly, $\hat{\beta} \sim \mathcal{N}(\tau + \delta, \Sigma)$ with within-cohort AR(1) covariance $\Sigma_{ij} = s^2 \rho^{|i-j|}$. The panel model draws individual untreated outcomes $Y_{it}(0) = \delta_g(e) + \epsilon_{it}$, $\epsilon \sim \mathcal{N}(0, \sigma_\epsilon^2)$, for 500 treated units per cohort and 1500 shared controls over $T = 14$ periods, and estimates the coefficients and their covariance from the data.

	Estimand (Table 3)	Scope (Fig. 1)	Honest cov. (Table 4, Fig. 2)	Panel (Sec. 3.5)
Cohorts (clean/confounded)	6 (4/2)	12 (6/6)	—	6 (3/3)
Post-treatment violation V	0.8	0.6	0.3, 0.6	0.6
Informativeness ϕ	1.0	$[0, 1.5]$	—	$[0, 1.5]$
Screen threshold c	0.40	0.40	—	0.40
Noise	s=0.03–0.40	σ=0.20	σ_{agg}=0.10	σ_ϵ=0.5
Pre/post correlation ρ	0.5	0.5	—	estimated
Replications	20,000	8,000	4,000	2000–3000

the ATT averages the post-treatment effect over all cohorts, as a heterogeneity-robust estimator would, and the LATT averages it over the selected cohorts only. Because the screen reads the selected set off the same noisy pre-trends, the selection is genuinely data-dependent in the sense of Section 2, and the pre-test bias of Proposition 4 is operative.

3.2 Recovering a clean effect where the ATT is biased

We first illustrate Proposition 1 in the simplest configuration, with a subset of cohorts violating parallel trends in the post-period. Table 3 reports the result. The estimator of the ATT is biased by +0.267, since it averages over all cohorts, so the offending cohorts' differential trends enter it directly. The LATT lands at 1.000, on the truth, because the selection removes those cohorts from the average and, by Proposition 1, their violations then do not enter the estimand at all.

The lower panel of the table sweeps the sampling noise s. As precision improves the LATT's bias falls monotonically from +0.013 to zero, and this residual is the pre-test bias of Proposition 4, which vanishes as the selection rule becomes reliable. Two sources could in principle generate it, a noisy screen that admits genuinely confounded cohorts, which would persist even under independent selection, and the correlation between a cohort's pre- and post-treatment sampling errors, which distorts the retained cohorts' estimates only under same-sample selection. The final column isolates the two by selecting on an independent pre-trend draw. It tracks the full-sample column almost exactly, so the residual is the first source, admission of confounded cohorts, and the second is negligible here because the two-sided flatness screen conditions the retained estimates symmetrically. The ATT estimator's bias, by contrast, is frozen at +0.267 regardless of precision. The LATT faces a noise problem, which more data cures, whereas the ATT estimator faces a wrong-target problem, which no amount of data cures.

Table 3: The estimand demonstration (reduced-form model). True effect = 1 for all cohorts. In the lower panel LATT (full) selects on the same pre-trends used for estimation, while LATT (split) selects on an independent draw. Their near-equality shows the residual bias is admission of confounded cohorts, not same-sample conditioning.

	ATT	LATT (full)	LATT (split)
Point estimate	1.267	1.000	
Bias vs. truth	+0.267	+0.000	
Residual LATT bias as sampling noise s falls			
$s = 0.40$	+0.265	+0.013	+0.014
$s = 0.20$	+0.267	−0.001	−0.000
$s = 0.12$	+0.267	−0.000	−0.000
$s = 0.03$	+0.267	+0.000	+0.000

3.3 The scope condition

Figure 1 traces the method across the informativeness axis ϕ described in Section 3.1.

The upper-left panel plots the bias of the two point estimators against the truth. The bias of the ATT estimator is a flat line at +0.30, since it never consults pre-trends, so its performance cannot depend on how informative they are. The LATT's bias begins on top of that line at $\phi = 0$ and falls to zero as pre-trends become informative. The vertical gap between the two curves is therefore exactly the method's advantage, and reading it off from left to right gives the scope condition, with the advantage nil when pre-trends carry no information about post-treatment violations and maximal when they carry a great deal.

That the two curves coincide at the left edge is not a defect but an honest reflection of the fact, argued in Section 2, that selecting on uninformative pre-trends buys nothing.

The upper-right panel explains why the LATT's bias falls, by plotting the identification gap of the selected set, the average post-treatment violation among the cohorts that survive selection. It tracks the LATT's bias almost exactly, falling from 0.30 to essentially zero. This confirms that the LATT's remaining bias is not an artifact of the estimator but the residual violation that selection has failed to remove. The lower-right panel gives the mechanism behind both, the average number of confounded cohorts that slip through the screen falling from 5.09 of six when the confound is invisible to essentially zero when it is plainly visible.

The lower-left panel turns to inference and reports the coverage of a point-based confidence interval for the causal target, computed both in the naive way and by sample-splitting. Both collapse at low informativeness and recover only at high informativeness. The comparison is diagnostic, since sample-splitting removes any contamination from data-dependent selection by construction, so the fact that it collapses just as badly establishes that the failure is not selection distortion but the identification gap of the upper-right panel, since the surviving cohorts genuinely still violate parallel trends. This is what motivates the sensitivity bounds of the next subsection, since no purely

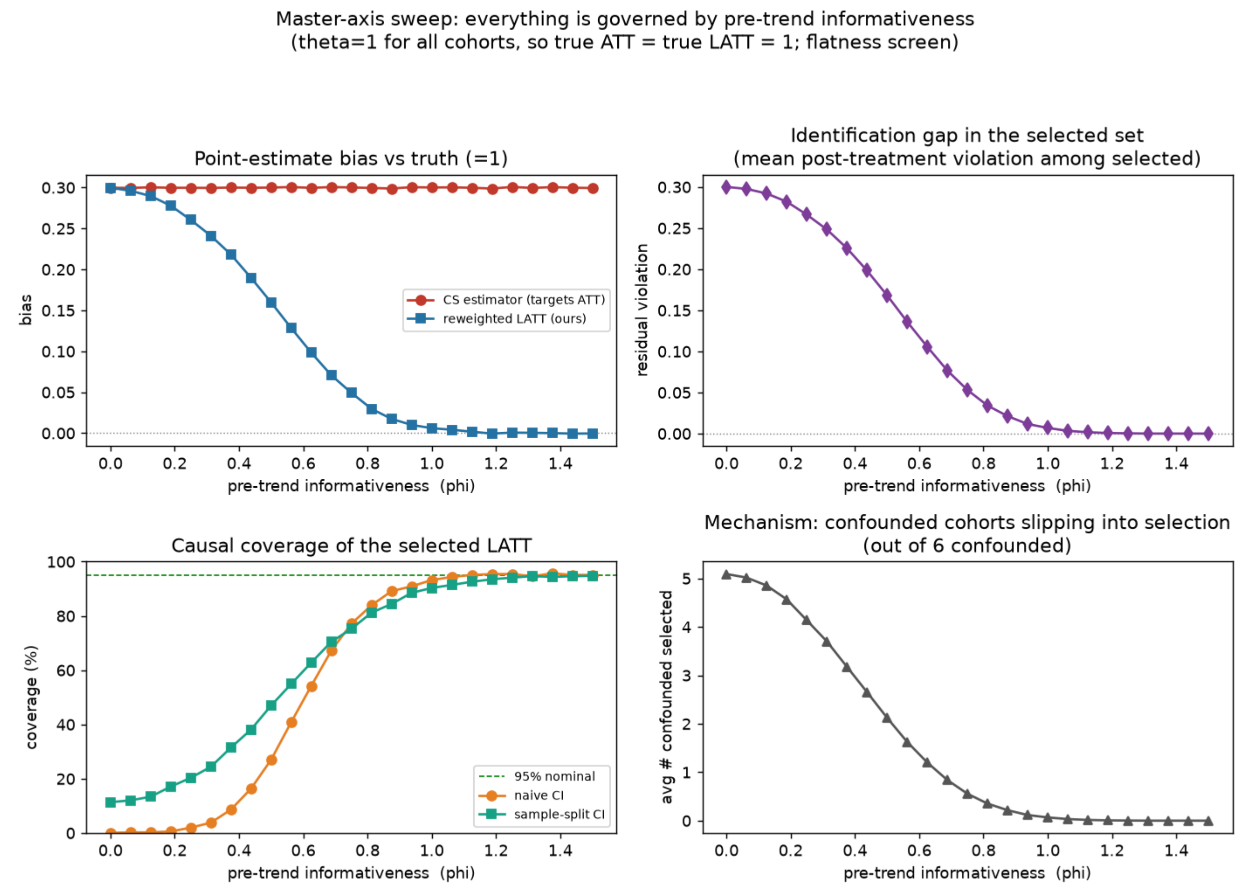

Figure 1: Scope condition. Everything is governed by pre-trend informativeness. The ATT estimator's bias is constant, and the LATT's advantage is the gap between the lines. Causal-coverage collapse at low informativeness is an identification gap (surviving cohorts still violate parallel trends), not selection distortion, since sample-splitting collapses just as badly.

inferential fix can repair a violation of the identifying assumption.

(At the right edge the naive interval slightly outperforms the split one. This is because splitting discards half the data, and by that point selection is nearly deterministic, so there is no contamination left to correct.)

3.4 Honest inference restores coverage

We now validate the level-bound FLCI of Section 2 applied to the selected-cohort aggregate, using the level violation of Section 3.1 whose size V is directly comparable to the assumed bound M.

Table 4 is the validity check. When the assumed bound equals the true violation ($V = M$) the FLCI covers at 95.0%, that is, at its nominal level. When the bound is generous ($M > V$) it covers conservatively at 100%. When the bound is violated ($M < V$) coverage collapses. This is the behavior an honest sensitivity interval should display. It is valid on the range of violations it claims to accommodate, and it fails visibly outside it, rather than failing silently.

Table 4: FLCI coverage of the causal target (reduced-form model), for an aggregate whose violation equals V, valid iff $M \geq V$.

True violation V	Assumed M	FLCI coverage	Point-estimate coverage
0.30	0.30 $(= V)$	95.0%	15.9%
0.30	0.60 $(> V)$	100%	14.9%
0.60	0.30 $(< V)$	9.3%	0.0%
0.60	0.60 $(= V)$	95.0%	0.0%

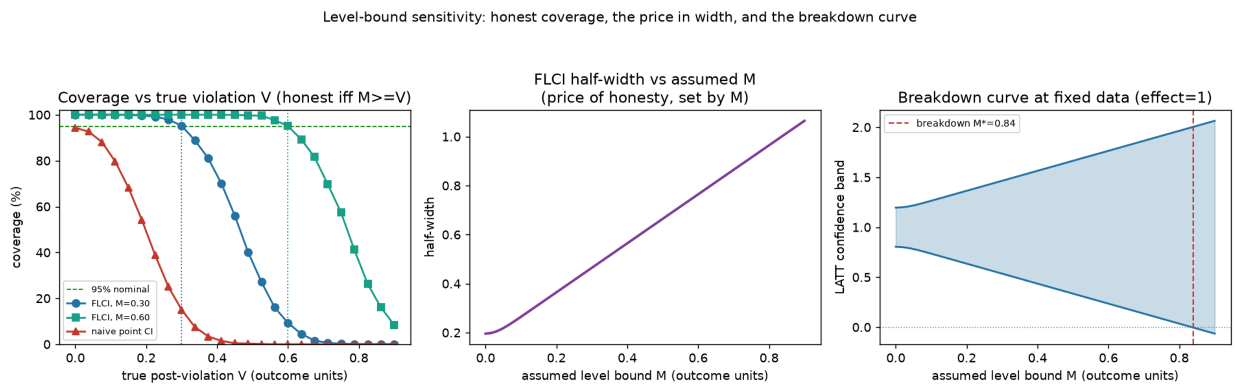

Figure 2: Sensitivity bounds restore honest coverage. The point estimate collapses as the violation grows. The FLCI is honest exactly while $M \geq V$. Interval width is the price of honesty, set by M. The right panel is the sensitivity/breakdown curve, with M^* in outcome units.

The final column records that the naive point interval collapses to near-zero coverage the instant any violation is present, because it is centered on a biased estimate with a width reflecting sampling noise alone.

Figure 2 repeats the exercise across a continuum of violation sizes and adds the practical output. In the left panel the point interval falls away rapidly as the violation grows, while each FLCI holds its nominal level until the true violation reaches its own assumed bound, marked by the vertical dotted lines, and declines thereafter. The middle panel records the price. Interval half-width is governed by the assumed M rather than by the realized violation, so honesty about a wider class of violations is paid for in width, immediately and proportionately.

The right panel is what a practitioner would actually report. Holding the data fixed, it traces the confidence band as the assumed bound is relaxed and identifies the breakdown value M^* at which the band first admits the truth. Because M is in the outcome's own units, this quantity answers the question a reader of a DiD paper wants answered directly, how large a post-treatment parallel-trends violation one must be willing to countenance before the stated conclusion no longer follows. And because the bound is fixed by the researcher rather than read off the pre-trends, the exercise remains informative even when pre-trends are uninformative, in direct contrast to the relative-magnitudes restriction, whose bounds would shrink toward zero in exactly that case and deliver false precision.

3.5 Selection, calibration, and panel confirmation

Two questions remain for the data-driven regime, whether choosing the selected set from the same data distorts the sensitivity interval, and how to choose M when the selected set, and hence its residual violation, is random. On the first, the cost of choosing the set from the same data is concentrated near the threshold and falls on the variance rather than on gross coverage. The M at which coverage crosses the nominal level is nearly the same for the full-data and sample-splitting procedures ($M \approx 0.048$ and 0.056), so where cohorts are well separated from c selection is effectively deterministic, naive inference suffices, and splitting buys little, consistent with the pointwise reading of Leeb and Pötscher (2005). That near-equality of crossing points understates the cost, however, because selection distorts the standard error rather than the crossing bound—the local-to-threshold regime the carving of Proposition 6 is built for, which the panel model below makes precise. On the second, the residual violation of the selected aggregate is not a constant but a spread, multi-modal random variable across replications, running from zero to about 0.06 with a thin tail to 0.11, each mode a different number of borderline cohorts surviving the screen, which is the content of Lemma 1. A fixed M must dominate it, and setting M at the mean residual violation (0.043) undercovers, while nominal coverage is reached near the ninety-fifth percentile (0.057) and well below the maximum (0.112). The recommendation follows, to calibrate M to the residual violation of a typical selected set, not its average, and treat it as a floor.

Finally, we repeat the estimand demonstration on generated panel data, where individual outcomes are simulated for a staggered design with a level confound and idiosyncratic noise, cohort event-study coefficients are estimated from genuine difference-in-differences contrasts against a never-treated group, and their covariance is estimated rather than supplied, which is the empirical counterpart of Remark 1. The conclusions are unchanged, since across the informativeness axis the ATT estimator's bias stays flat near $+0.30$ while the LATT's falls toward zero, reproducing Figure 1 with estimated quantities, and the estimated standard error matches its Monte Carlo counterpart where selection is effectively deterministic. One discrepancy is informative, and it is the payoff promised above. At intermediate informativeness, where selection is stochastic—the local-to-threshold regime the carved interval targets—the estimated standard error understates its Monte Carlo counterpart (0.016 against 0.024 at $\phi = 0.75$), the missing component being the variance induced by the randomness of selection, which a standard error conditional on the realized set does not capture. This confirms, from the panel side, that conditioning on a data-dependent selection understates uncertainty, and why the honest object to report is the sensitivity interval of Section 3.4 rather than a conventional confidence interval.

4 Empirical application

We illustrate the method with the local effect of the shale boom on house prices, a staggered design with many counties per cohort. The treatment date is the year a county's oil and gas production first rises sharply, from county production records (U.S. Department of Agriculture, Economic Research

Service, 2015), which sorts the boom counties into onset cohorts, and the outcome is the log county house-price index (Bogin et al., 2019), over 1998–2019. We estimate Callaway and Sant'Anna (2021) group-time effects against never-treated counties, those with negligible production, screen cohorts on the flatness of their estimated pre-trend, and report the level bound of Section 2 for the selected aggregate.

Onset is gradual and partly endogenous, which is what makes the setting instructive. A county's production ramps up over several years, and where it does so during a regional housing upswing its prices are already climbing before the boom is dated. Figure 3(a) shows the result. The early cohorts have flat pre-trends and are retained, while the later cohorts, whose onset coincides with the mid-2000s house-price run-up, carry pre-treatment trends several times larger and are dropped.

The two aggregates part ways (Figure 3b). Pooled across all cohorts, the event study climbs through the onset date to a house-price effect of +6.7 log points ($t = 8$), the kind of estimate a practitioner would report as a clear positive effect of the boom. But the pooled path is already trending before treatment, and the effect is inherited from the dropped cohorts. Restricted to the credible subpopulation, the three cohorts whose pre-trends are essentially flat (the screen at $c \approx 0.03$), the event study is flat both before and after onset, and the LATT is -0.2 log points, indistinguishable from zero. The gap between the two, +6.9 log points, is itself sharply estimated ($t = 5$). Honest inference confirms the reading rather than resting on the point estimate. Applying the level bound of Section 2 to the credible aggregate at M equal to the screen tolerance $c \approx 0.03$, the largest residual post-treatment violation the screen leaves room for, the LATT's fixed-length interval runs from -5.9 to $+5.5$ log points, which contains zero and excludes the pooled +6.7. The breakdown value is $M^* \approx 4.2$ log points, so reconciling the credible near-zero reading with the pooled effect would require countenancing a post-treatment violation about 1.4 times the screen tolerance and larger than any retained cohort's own pre-trend. The near-zero reading is not an artifact of a hand-picked threshold. Panel (c) traces the estimate along the credibility path of Section 2.3, and it stays flat and close to zero across the whole range of genuinely flat pre-trends, cohorts with $\max_e |\hat{\beta}_{g,\text{pre}}(e)|$ below about 0.03, and climbs toward the pooled +6.7 only once the threshold is loosened enough to readmit cohorts whose own pre-trends, around 0.07–0.08, are as large as the effect in question. Reading the LATT at that lax end simply rebuilds the pooled estimate out of its least credible cohorts, and the positive figure survives only by abandoning the credibility standard, not by tolerating a residual violation of the credible set.

The reading is that the pooled effect reflects a pre-existing trend in the counties that adopt late rather than a credibly causal effect, and a near-zero net effect for the credible subpopulation is what the offsetting amenity and disamenity channels of local energy development would predict (Muehlenbachs et al., 2015). The method identifies the effect for the credible cohorts alone, and the excess in the dropped cohorts blends their steep pre-trends with any genuine difference in their treatment effects, a mixture the design cannot separate and need not, since the credible reading stands on the retained cohorts. The application thus shows the method working in the direction opposite to point-identification, not recovering an effect the pooled estimate misses but withdrawing

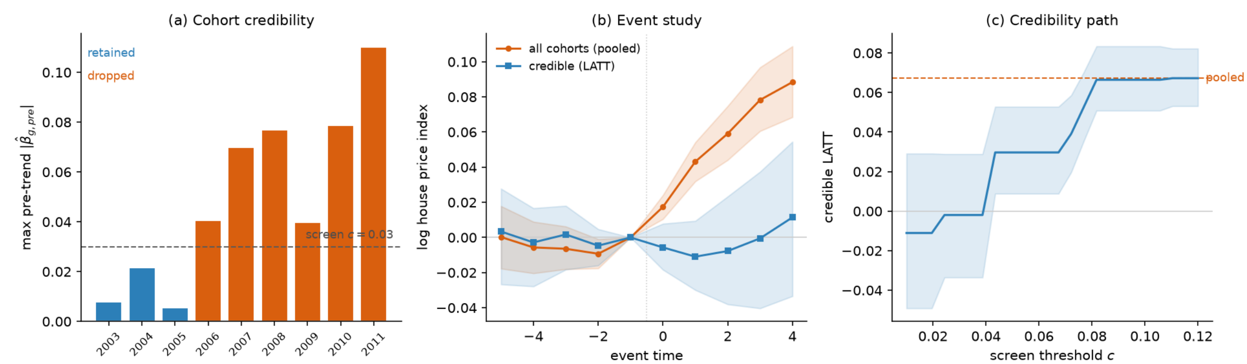

Figure 3: The shale boom and county house prices. (a) Cohort credibility map, the maximum pre-treatment violation by onset cohort with the flatness screen, where the later, trending cohorts are dropped. (b) Event study, all cohorts (pooled estimate) versus the credible subpopulation (LATT) at the illustrative threshold $c \approx 0.03$, with 95% bands. (c) The credibility path, the LATT as the screen threshold c is relaxed, near zero across the flat-pre-trend cohorts and rising toward the pooled estimate only as trending cohorts are readmitted.

one it spuriously reports, and locating that withdrawal exactly in the cohorts the screen flags.

5 Conclusion

We have argued that when parallel trends fails for some treated cohorts, a productive response is to change the estimand rather than to defend or bound the ATT. The credible-subpopulation LATT is point-identified under parallel trends for the selected cohorts alone, a condition that can hold when the ATT's fails, is estimated by a transparent reweighting of standard group-time effects, and can be accompanied by honest sensitivity bounds on the residual violation that the pre-trend screen cannot rule out. This is a specific instance of the general principle, familiar from instrumental variables and limited overlap, of retreating to a credibly-identified subpopulation.

The method delivers a sharp answer where the ATT estimator is biased and its sensitivity set is wide, at the price of a narrower, subpopulation target whose composition must be described rather than assumed. Its advantage is contingent on pre-trends being informative about post-treatment violations. Where they are not, only economic context can carry the assumption. Under data-driven selection the estimator carries a pre-test bias that vanishes as pre-trends grow informative; the inference we recommend is honest regardless, combining the post-selection carving of Lee et al. (2016) with the sensitivity bounds of Rambachan and Roth (2023) into an interval uniformly valid against both the selection and the residual violation, so that the pre-test problem the selection revives is met head-on rather than assumed away. We also make the scope condition operational: a feasible rule, reported alongside both honest intervals, signals when narrowing the target lowers risk, with a breakdown value Γ^* standing in for the one quantity—the effects of the discarded cohorts—that the data cannot identify.

An appendix characterizes the optimal credible weighting (Proposition 9): minimizing worst-case risk yields a soft-thresholded weighting of cohorts by credibility net of noise, of which the flatness screen with size weights is a special case, which completes the analogy to the variance-minimizing overlap weights of Crump et al. (2009) that the approach borrows from.

Two questions we characterize empirically and leave to future theory. The first is the finite-sample behaviour of the feasible optimal weighting, in which the credibility levels m_g are estimated from the pre-trends rather than known: Appendix A reports its typical gain in simulation, and a full account of the noise it inherits from the pre-trends is open. The second is the finite-sample coverage of a *single* interval chosen by a hard-thresholded switching rule; the both-intervals report we recommend is honest by Proposition 8 regardless of the choice, and Section 3.5 shows the selection itself introduces no gross coverage distortion, so what remains open is only the coverage of the single reported interval near the switching boundary.

A Optimal credible weighting

The flatness screen and the cohort weights w_g have so far been taken as given, the screen retaining a cohort outright and the weights fixed at, say, cohort size. Both are choices, and the sensitivity class that governs the honest inference also pins down their optimal form, in the decision-theoretic sense in which Crump et al. (2009) derive the optimal-overlap subpopulation and Li et al. (2018) the overlap weights: there, variance is minimized where positivity is scarce; here, worst-case mean squared error is minimized where credibility is scarce.

Let the estimator be a weighted average $\hat{\theta}(\lambda) = \sum_g \lambda_g \hat{\beta}_{g,\text{post}}$ with $\lambda_g \geq 0$ and $\sum_g \lambda_g = 1$, and—following the overlap-weighting logic that the weights define the estimand as well as the estimator—let the target be the λ-weighted causal effect $\theta(\lambda) = \sum_g \lambda_g \theta_g$, whose interpretation is reported through the weights themselves. Since $\hat{\beta}_{g,\text{post}} \sim \mathcal{N}(\theta_g + V_g, \sigma_g^2)$ independently, the estimator has identification bias $\sum_g \lambda_g V_g$ and variance $\sum_g \lambda_g^2 \sigma_g^2$. The pre-trends enter through the sensitivity class: a cohort's post-treatment violation is bounded by a credibility level $m_g \geq 0$, $|V_g| \leq m_g$, with m_g small for a cohort whose pre-trend is flat and large for one whose pre-trend is steep. The hard flatness screen is the coarsening $m_g \in \{0, \infty\}$, retaining the $m_g = 0$ cohorts and discarding the rest; the level bound M of Section 2.4 is the common value $m_g \equiv M$ on the retained set. The worst-case mean squared error over this class is

$$\mathcal{R}(\lambda) = \Big(\sum_g \lambda_g m_g\Big)^2 + \sum_g \lambda_g^2 \sigma_g^2,$$

the squared worst-case bias plus the variance, and the optimal credible weighting minimizes it over the simplex.

Proposition 9 (Optimal credible weighting). *$\mathcal{R}$ is strictly convex on the simplex, so it has a unique minimizer λ^*. There are scalars μ and $B = \sum_g \lambda_g^* m_g$, the worst-case bias at the optimum, such that*

$$\lambda_g^* = \frac{1}{\sigma_g^2}\Big[\tfrac{\mu}{2} - B\, m_g\Big]_+, \qquad [x]_+ = \max(x, 0),$$

with μ fixed by $\sum_g \lambda_g^* = 1$. Consequently there is an endogenous credibility cutoff $m^* = \mu/(2B)$: cohorts with $m_g \geq m^*$ receive zero weight, and among the retained cohorts the weight is $\lambda_g^* \propto (m^* - m_g)/\sigma_g^2$, decreasing in the credibility bound m_g and in the sampling variance σ_g^2.

Proof. $\sum_g \lambda_g^2 \sigma_g^2$ is strictly convex for $\sigma_g^2 > 0$ and $(\sum_g \lambda_g m_g)^2$ is convex, so $\mathcal{R}$ is strictly convex and its minimizer over the compact convex simplex is unique. The worst-case bias is $\sup_{|V_g| \leq m_g} |\sum_g \lambda_g V_g| = \sum_g \lambda_g m_g$ since $\lambda_g \geq 0$, giving the displayed $\mathcal{R}$. Form the Lagrangian $\mathcal{R}(\lambda) - \mu(\sum_g \lambda_g - 1) - \sum_g \nu_g \lambda_g$ with $\nu_g \geq 0$; stationarity gives $2Bm_g + 2\lambda_g \sigma_g^2 - \mu - \nu_g = 0$ with $B = \sum_g \lambda_g m_g$. For $\lambda_g > 0$ complementary slackness sets $\nu_g = 0$ and $\lambda_g = (\mu/2 - Bm_g)/\sigma_g^2$; for $\lambda_g = 0$ it requires $\mu/2 - Bm_g \leq 0$, i.e. $m_g \geq \mu/(2B)$. Combining the two cases gives the truncated form and the cutoff $m^* = \mu/(2B)$, with μ determined by the budget constraint and B by its own definition at the resulting λ^*. □

Three readings connect the result to the rest of the paper. First, the optimal rule is a *soft* flatness screen. It reproduces the hard screen's qualitative content—cohorts whose pre-trends are too steep are dropped—but with an endogenous threshold set by the bias–variance trade-off rather than a researcher's c, and with the surviving cohorts weighted continuously by how credible and how precise they are, rather than equally. A marginally less credible but large and precisely estimated cohort is retained where the hard screen would discard it, because the variance its inclusion saves outweighs the bias it admits—the same calculus as the dominance condition of Proposition 7, now internal to the weighting. Second, the result justifies the paper's default. Among equally credible cohorts, $m_g \equiv 0$, the weights collapse to inverse-variance, $\lambda_g^* \propto 1/\sigma_g^2$, and when sampling variance scales inversely with cohort size, $\sigma_g^2 \propto 1/n_g$, inverse-variance weighting is size weighting; the fixed w_g of the definition is thus optimal precisely under homoskedastic-per-unit noise and equal credibility, and the proposition says what to do when either fails. Third, it completes the analogy to limited overlap. Crump et al. (2009) trim and Li et al. (2018) weight to minimize variance where positivity is the binding scarcity; here credibility is the scarcity, and the optimal response is to weight cohorts by credibility net of noise, with trimming emerging as the corner solution for cohorts past m^*.

The credibility levels m_g are reported, not assumed, in the same currency as everything else: a natural choice sets m_g to the cohort's own maximal pre-trend, $m_g = \max_{e<0}|\hat{\beta}_{g,\text{pre}}(e)|$, so that the weighting is read off the observed pre-trends. This makes the weights data-driven, a refinement of the selection already treated, and—as there—validity does not rest on it: the honest interval of Section 2.4 takes its coverage from the researcher-set level bound M, while the estimated m_g only orders and weights the cohorts. Using m_g to weight but M to cover keeps efficiency and honesty on separate instruments, as the carved inference of Proposition 6 keeps selection and identification on separate instruments.

A.1 Simulation: the optimal weighting improves on the hard screen

Proposition 9 predicts that the soft, credibility-and-precision weighting lowers risk relative to the hard flatness screen. We confirm this and measure the gain. The design has $K = 12$ cohorts with homogeneous effects, so every weighting targets the same value and mean squared error is a fair comparison. Cohorts come in pairs at six violation levels $V_g \in \{0, 0.08, \ldots, 0.40\}$; within each pair one member is precise (a large sample) and one noisy (a small one). A hard screen keeps or drops a pair as a whole, since both members share the credibility m_g, whereas the optimal weighting can separate them. The design is deliberately constructed to exhibit this precision channel, and a random-DGP average below reports the typical gain.

We compare three procedures for the common target: the ATT, equal weights on all cohorts; the hard screen, equal weights on $\{g : m_g \leq c\}$ at its best threshold c; and the optimal weighting λ^* of Proposition 9, in an oracle version with $m_g = V_g$ known and a feasible version with $\hat{m}_g = \max_e|\hat{\beta}_{g,\text{pre}}(e)|$ estimated from informative pre-trends.

Table 5 and Figure 4 report the result. The left panel plots MSE against the hard-screen threshold: the optimal weighting lies below the hard-screen curve at every c, including its minimum.

Table 5: Optimal weighting versus the hard screen (reduced-form model, homogeneous effects, so all procedures target the same value). RMSE and MSE for that target; the optimal weighting minimizes worst-case MSE by Proposition 9.

Procedure	RMSE	MSE
ATT (all cohorts)	0.200	0.0402
Hard screen (best c)	0.032	0.00104
Optimal weighting, feasible ($\hat{m}_g$)	0.022	0.00048
Optimal weighting, oracle ($m_g = V_g$)	0.020	0.00039

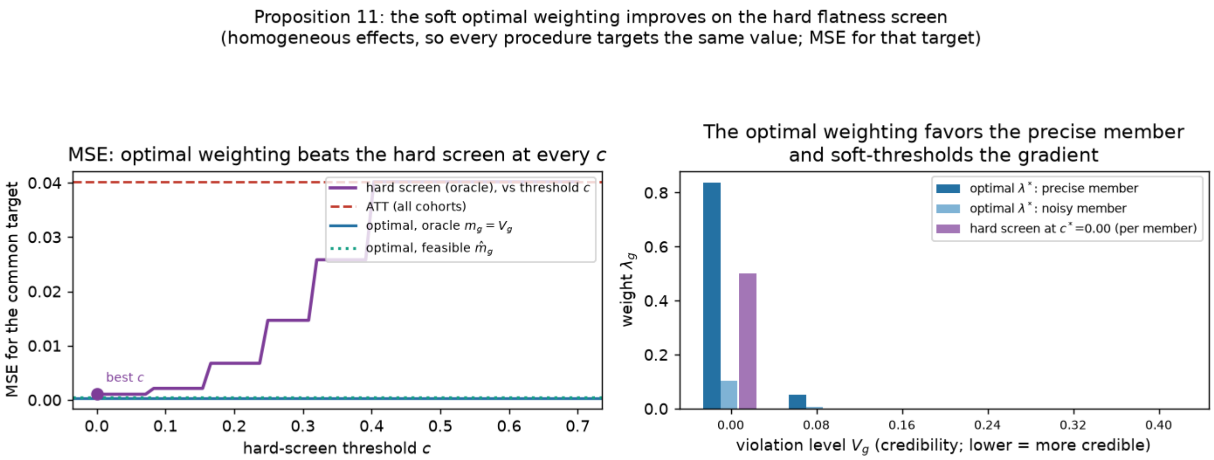

Figure 4: Optimal credible weighting (Proposition 9) versus the hard flatness screen, homogeneous effects so every procedure targets the same value. Left: MSE against the hard-screen threshold c; the optimal weighting, oracle and feasible, lies below the hard-screen curve at every c and below the ATT. Right: weights at each violation level; the optimal weighting loads on the precise member of each pair and soft-thresholds the credibility gradient, where the hard screen weights retained cohorts equally.

Against the best hard screen, the optimal weighting lowers MSE by 63% when credibility is known. The feasible version, which estimates m_g from the pre-trends and is compared to the best hard screen doing the same, lowers it by 28%; across 500 random DGPs the average reduction over the best hard screen is 14%. The right panel shows the mechanism. At a common credibility level the optimal weighting loads on the precise member and down-weights the noisy one, where the hard screen weights them equally; and it retains marginally less credible cohorts at reduced weight rather than dropping them at a cutoff. The gap between the oracle and feasible lines is the price of estimating m_g from noisy pre-trends, the open question the conclusion notes; even so, the feasible weighting improves on the hard screen.

References

Abadie, Alberto, Alexis Diamond, and Jens Hainmueller, "Synthetic Control Methods for Comparative Case Studies: Estimating the Effect of California's Tobacco Control Program," *Journal of the American Statistical Association*, 2010, *105* (490), 493–505.

Armstrong, Timothy B. and Michal Kolesár, "Optimal Inference in a Class of Regression Models," *Econometrica*, 2018, *86* (2), 655–683.

Bilinski, Alyssa and Laura A. Hatfield, "Nothing to See Here? Non-inferiority Approaches to Parallel Trends and Other Model Assumptions," *arXiv preprint arXiv:1805.03273*, 2020.

Bogin, Alexander N., William M. Doerner, and William D. Larson, "Local House Price Dynamics: New Indices and Stylized Facts," *Real Estate Economics*, 2019, *47* (2), 365–398.

Borusyak, Kirill, Xavier Jaravel, and Jann Spiess, "Revisiting Event-Study Designs: Robust and Efficient Estimation," *Review of Economic Studies*, 2024. Forthcoming.

Callaway, Brantly and Pedro H. C. Sant'Anna, "Difference-in-Differences with Multiple Time Periods," *Journal of Econometrics*, 2021, *225* (2), 200–230.

Crump, Richard K., V. Joseph Hotz, Guido W. Imbens, and Oscar A. Mitnik, "Dealing with Limited Overlap in Estimation of Average Treatment Effects," *Biometrika*, 2009, *96* (1), 187–199.

de Chaisemartin, Clément, "Using Pre-Trends for Inference in Difference-in-Differences," *arXiv preprint arXiv:2607.21312*, 2026.

_ **and Xavier D'Haultfœuille**, "Two-Way Fixed Effects Estimators with Heterogeneous Treatment Effects," *American Economic Review*, 2020, *110* (9), 2964–2996.

Freyaldenhoven, Simon, Christian Hansen, and Jesse M. Shapiro, "Pre-event Trends in the Panel Event-Study Design," *American Economic Review*, 2019, *109* (9), 3307–3338.

Goodman-Bacon, Andrew, "Difference-in-Differences with Variation in Treatment Timing," *Journal of Econometrics*, 2021, *225* (2), 254–277.

Imbens, Guido W. and Joshua D. Angrist, "Identification and Estimation of Local Average Treatment Effects," *Econometrica*, 1994, *62* (2), 467–475.

Kahn-Lang, Ariella and Kevin Lang, "The Promise and Pitfalls of Differences-in-Differences: Reflections on *16 and Pregnant* and Other Applications," *Journal of Business and Economic Statistics*, 2020, *38* (3), 613–620.

Kwon, Soonwoo and Jonathan Roth, "(Empirical) Bayes Approaches to Parallel Trends," *AEA Papers and Proceedings*, 2024, *114*, 606–609.

Lee, Jason D., Dennis L. Sun, Yuekai Sun, and Jonathan E. Taylor, "Exact Post-Selection Inference, with Application to the Lasso," *The Annals of Statistics*, 2016, *44* (3), 907–927.

Leeb, Hannes and Benedikt M. Pötscher, "Model Selection and Inference: Facts and Fiction," *Econometric Theory*, 2005, *21* (1), 21–59.

_ **and** _ , "Sparse Estimators and the Oracle Property, or the Return of Hodges' Estimator," *Journal of Econometrics*, 2008, *142* (1), 201–211.

Li, Fan, Kari Lock Morgan, and Alan M. Zaslavsky, "Balancing Covariates via Propensity Score Weighting," *Journal of the American Statistical Association*, 2018, *113* (521), 390–400.

Manski, Charles F. and John V. Pepper, "How Do Right-to-Carry Laws Affect Crime Rates? Coping with Ambiguity Using Bounded-Variation Assumptions," *Review of Economics and Statistics*, 2018, *100* (2), 232–244.

Muehlenbachs, Lucija, Elisheba Spiller, and Christopher Timmins, "The Housing Market Impacts of Shale Gas Development," *American Economic Review*, 2015, *105* (12), 3633–3659.

Rambachan, Ashesh and Jonathan Roth, "A More Credible Approach to Parallel Trends," *Review of Economic Studies*, 2023, *90* (5), 2555–2591.

Roth, Jonathan, "Pretest with Caution: Event-Study Estimates after Testing for Parallel Trends," *American Economic Review: Insights*, 2022, *4* (3), 305–322.

_ **, Pedro H. C. Sant'Anna, Alyssa Bilinski, and John Poe**, "What's Trending in Difference-in-Differences? A Synthesis of the Recent Econometrics Literature," *Journal of Econometrics*, 2023, *235* (2), 2218–2244.

Sun, Liyang and Sarah Abraham, "Estimating Dynamic Treatment Effects in Event Studies with Heterogeneous Treatment Effects," *Journal of Econometrics*, 2021, *225* (2), 175–199.

U.S. Department of Agriculture, Economic Research Service, "County-level Oil and Gas Production in the United States," 2015. Onshore production, lower 48 states, 2000–2011.